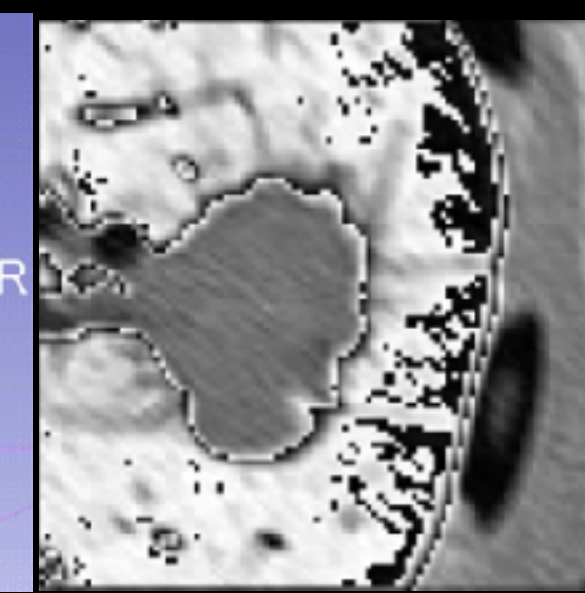
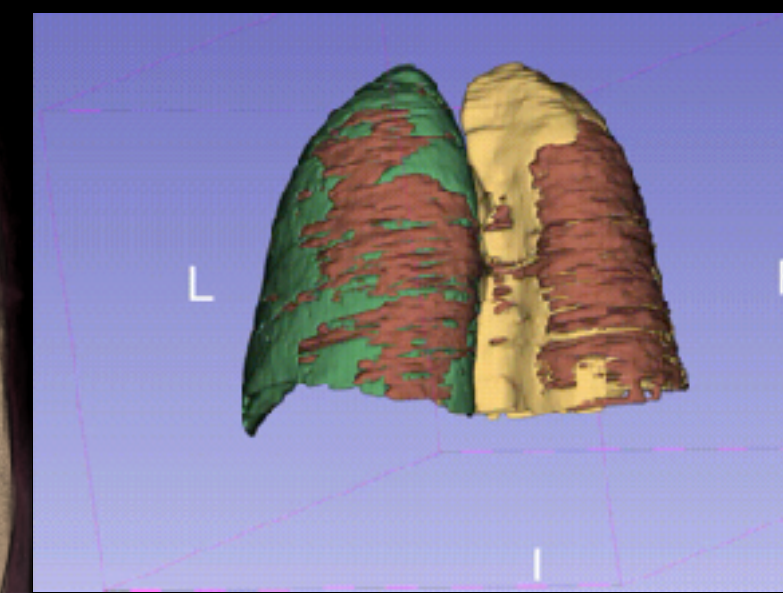
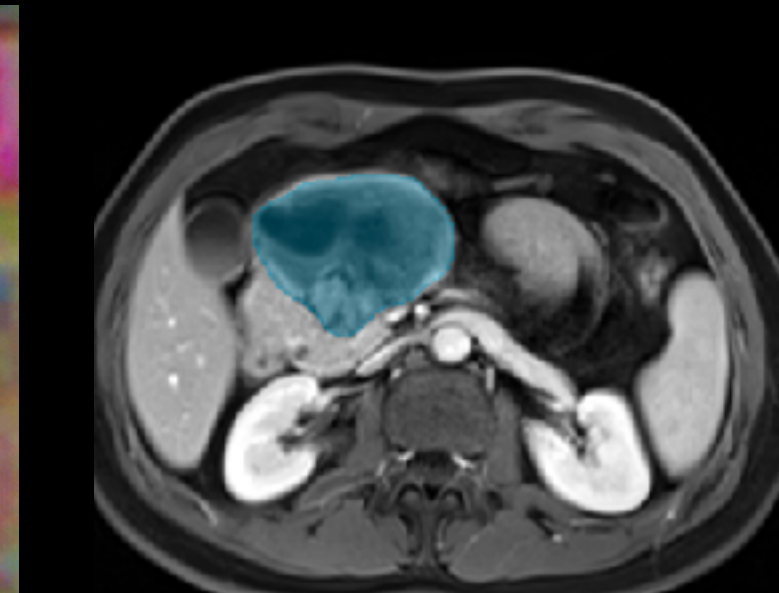
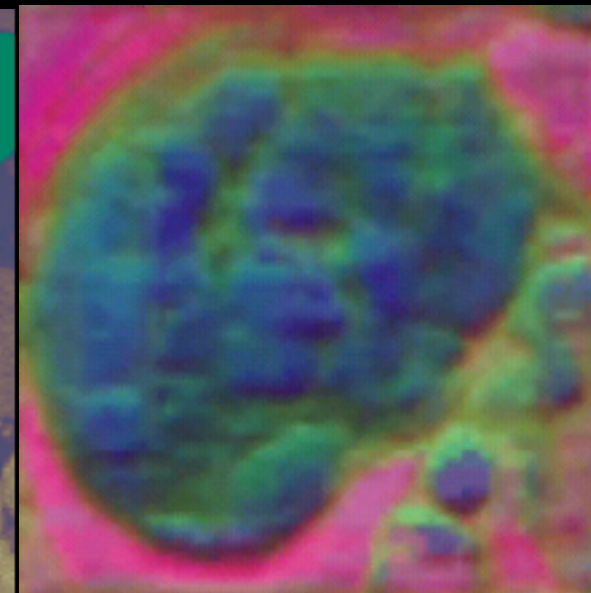
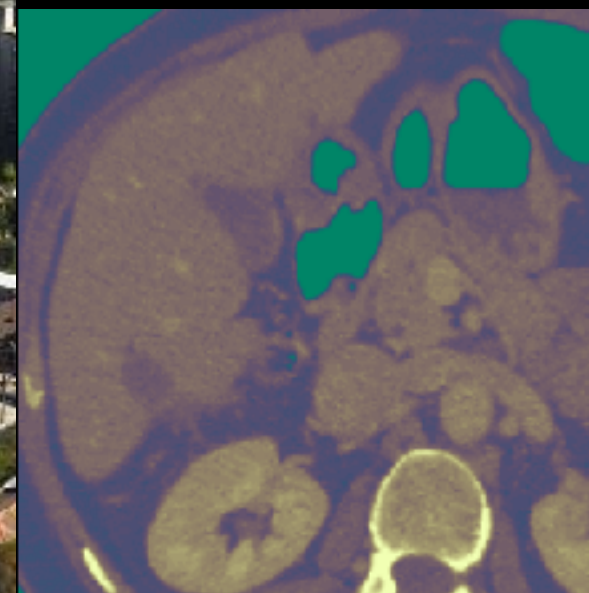




Generalidades de Machine Learning

Curso de Investigación en Radiología



Héctor Henríquez Leighton MD, MS
hhenriquez@miuandes.cl
Profesor Asistente Universidad de los Andes
Investigador Principal Computer Vision Lab
Clínica Santa María, Santiago de Chile



Universidad de
los Andes

Contenidos

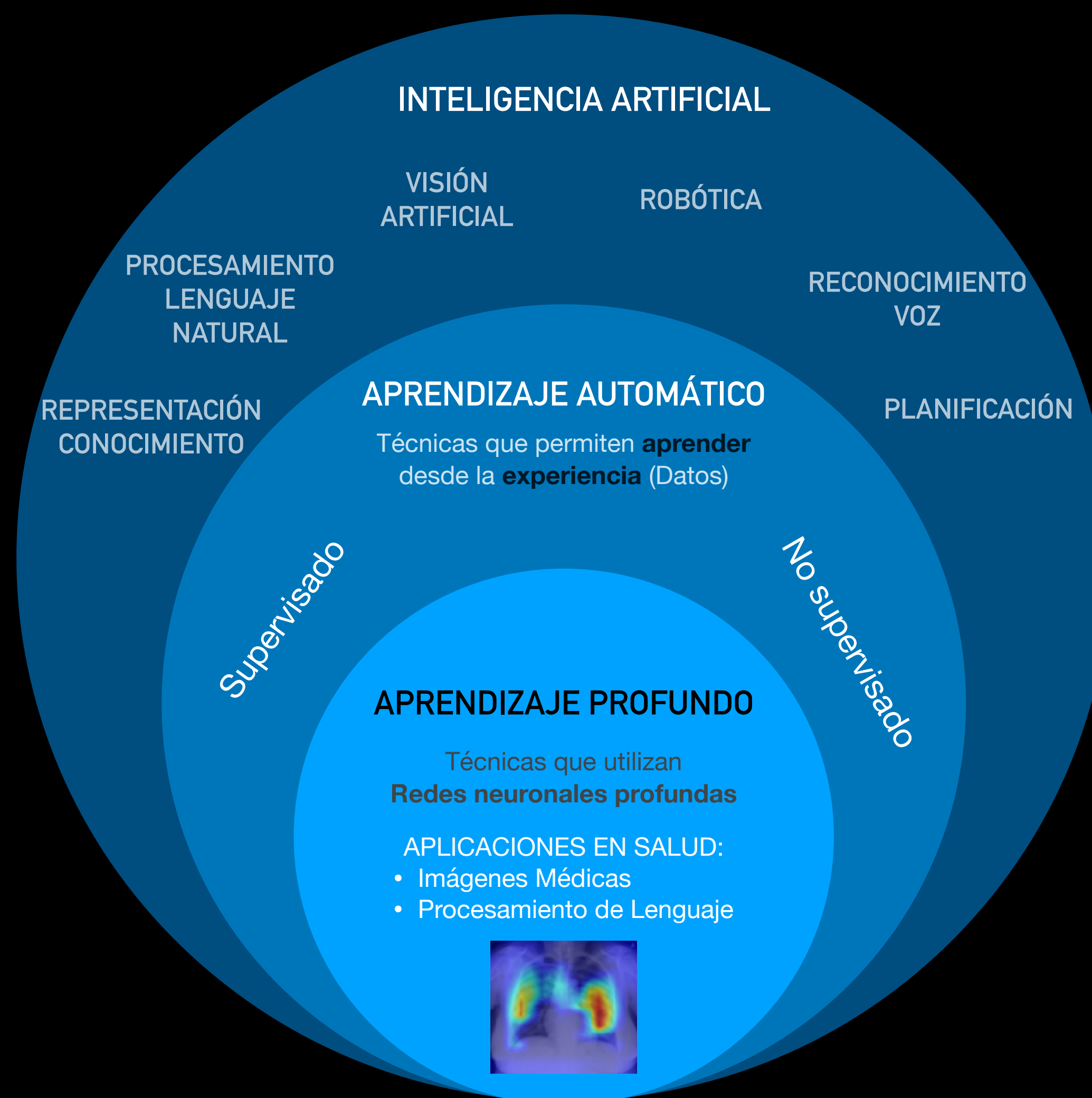
- Generalidades de Machine Learning.
- Proceso de anotación.
- Modelos de Visión por Computadora.

No tengo conflictos de interés

¿Dónde estoy?

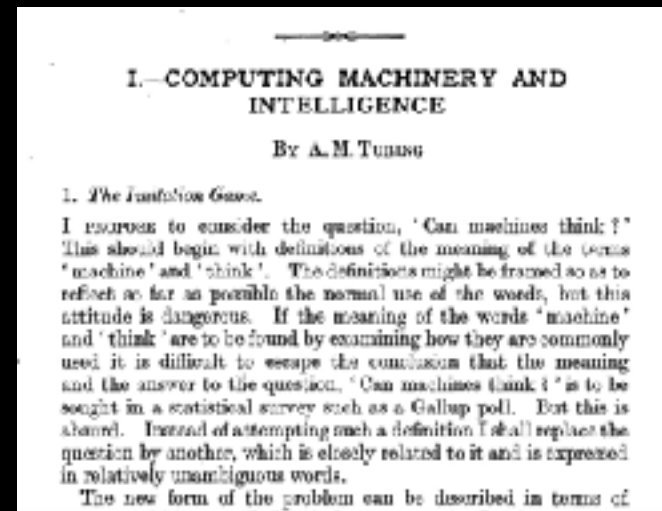


Inteligencia Artificial



Algunos hitos en la historia de la IA

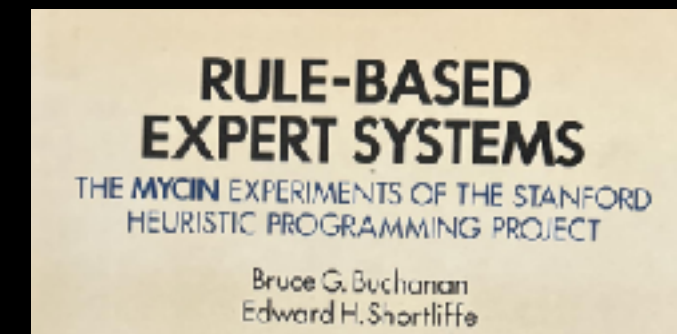
Alan Turing
"Imitation Game"



Frank Rosenblat
Perceptron Mark I



MYCIN
Ted Shortliffe



IBM Deep Blue
vs
G. Kasparov



1936

1956

1966

1989

1950

1957

1972

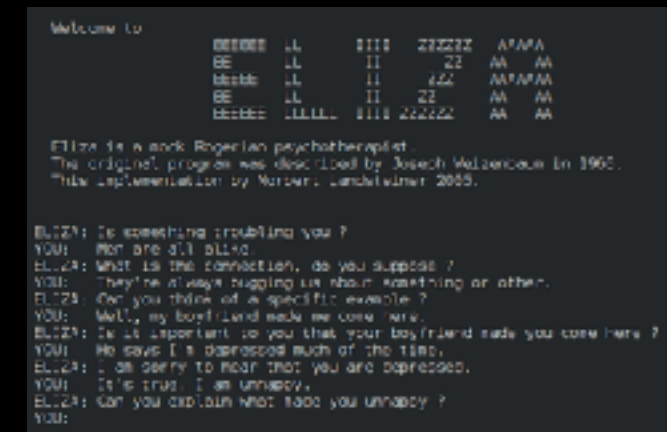
1997



Alan Turing
Formalización
Algoritmo



John McCarthy
Conferencia Darmouth



ELIZA
Joseph Weizenbaum



LeNet
Yann LeCun

Paradigma simbólico

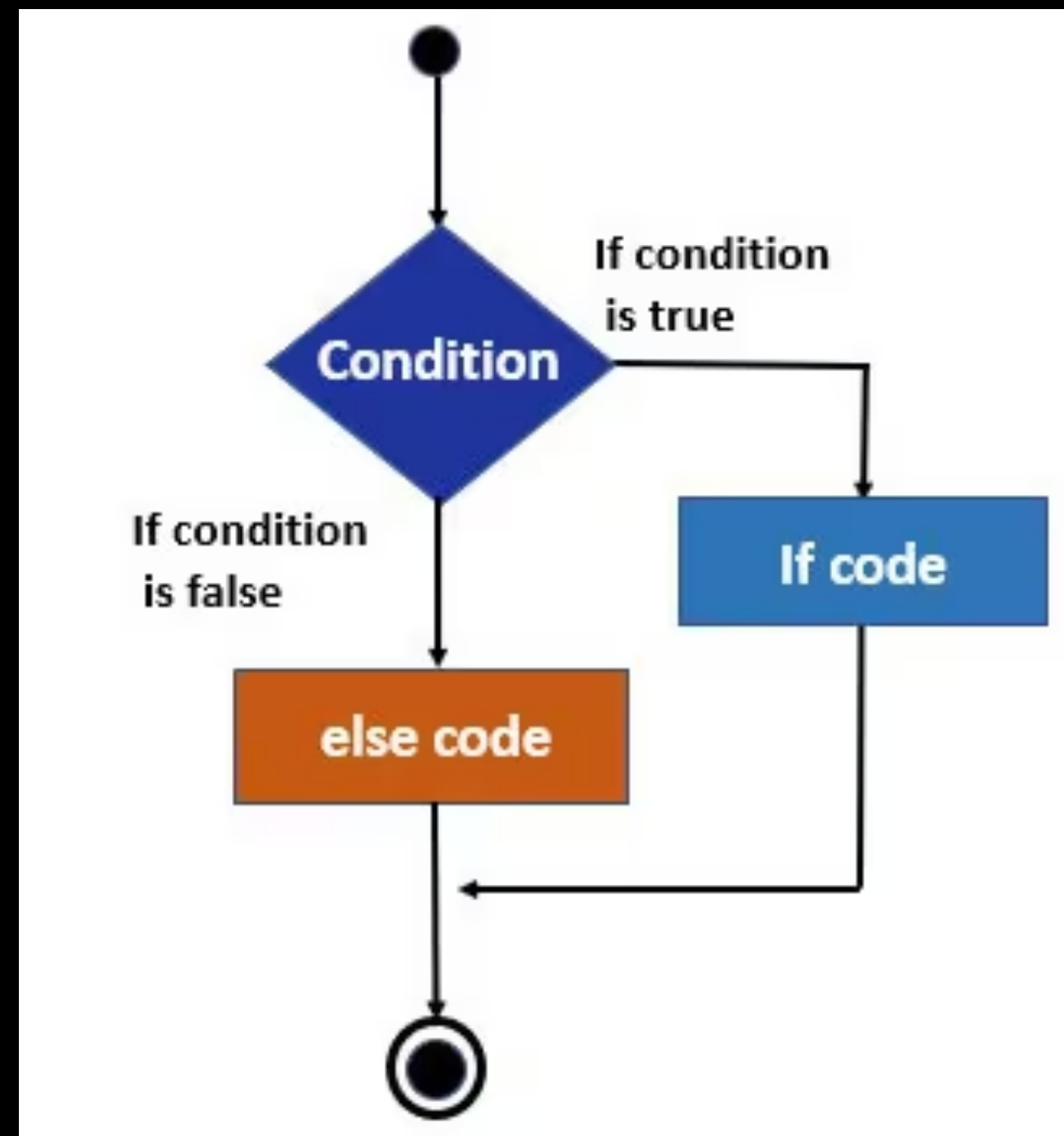


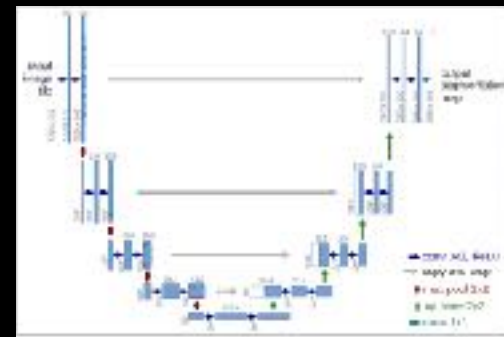
Imagen generada con Sora - OpenAI

Algunos hitos en la historia de la IA

AlexNet
Krizhevsky, Sutskever
& Hinton



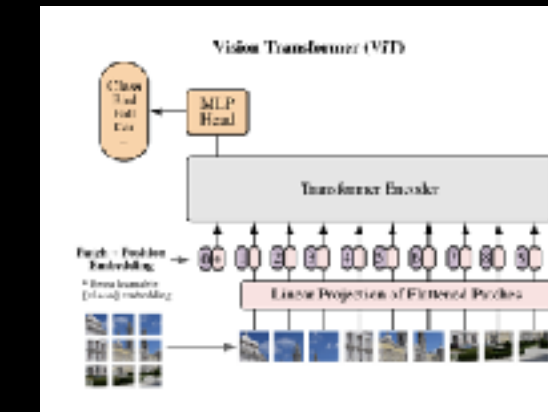
U-Net
Ronneberg et al



Transformers
Google



Vision Transformer
Dosovitsky



ChatGPT OpenAI



2012



ResNet
Ganador Imagenet
<5% de error

2015



AlphaGo
DeepMind
Vs
Lee Sedol

2017



PLM / LLM

2020



DALL-E OpenAI

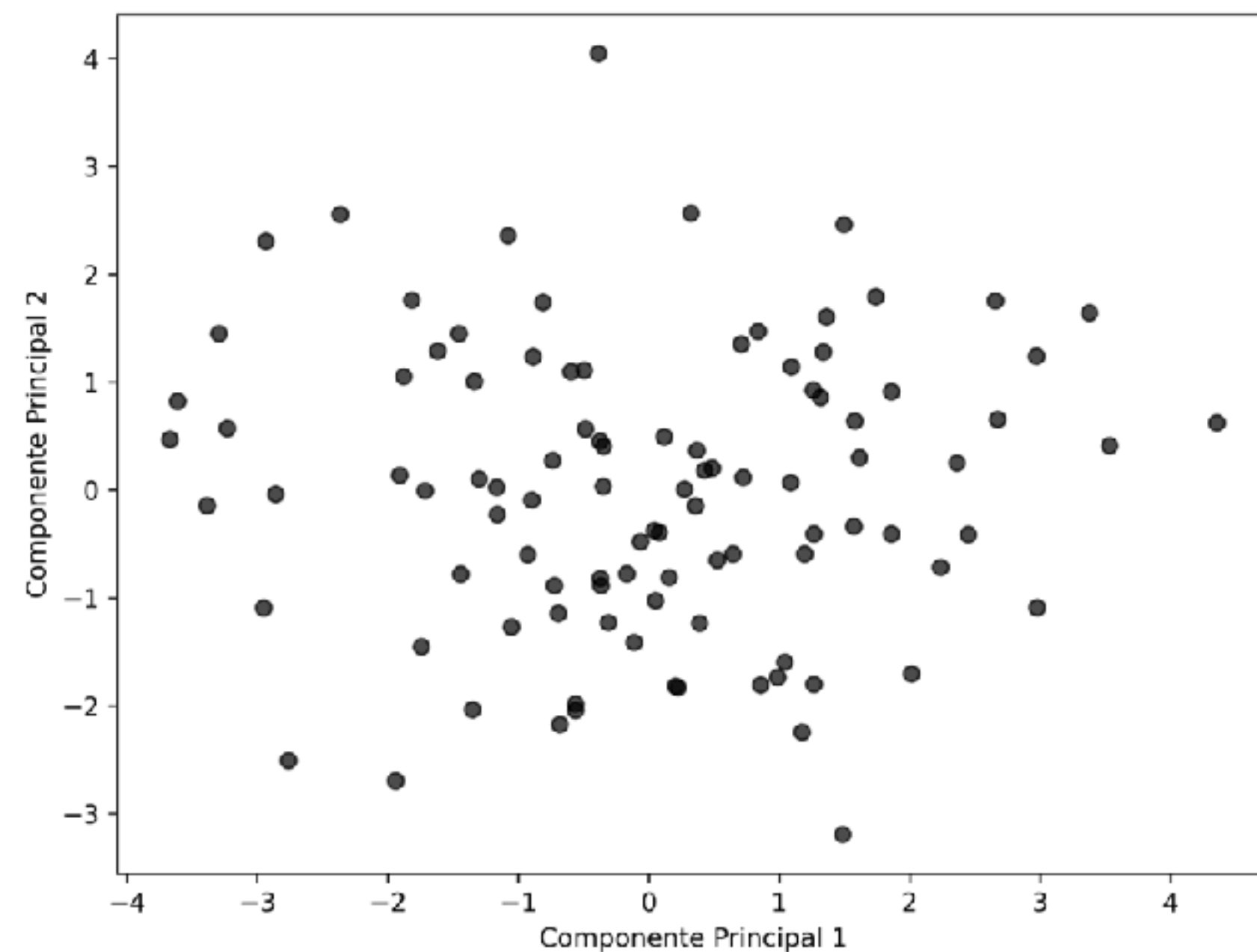
2021

2022

Paradigma subsimbólico

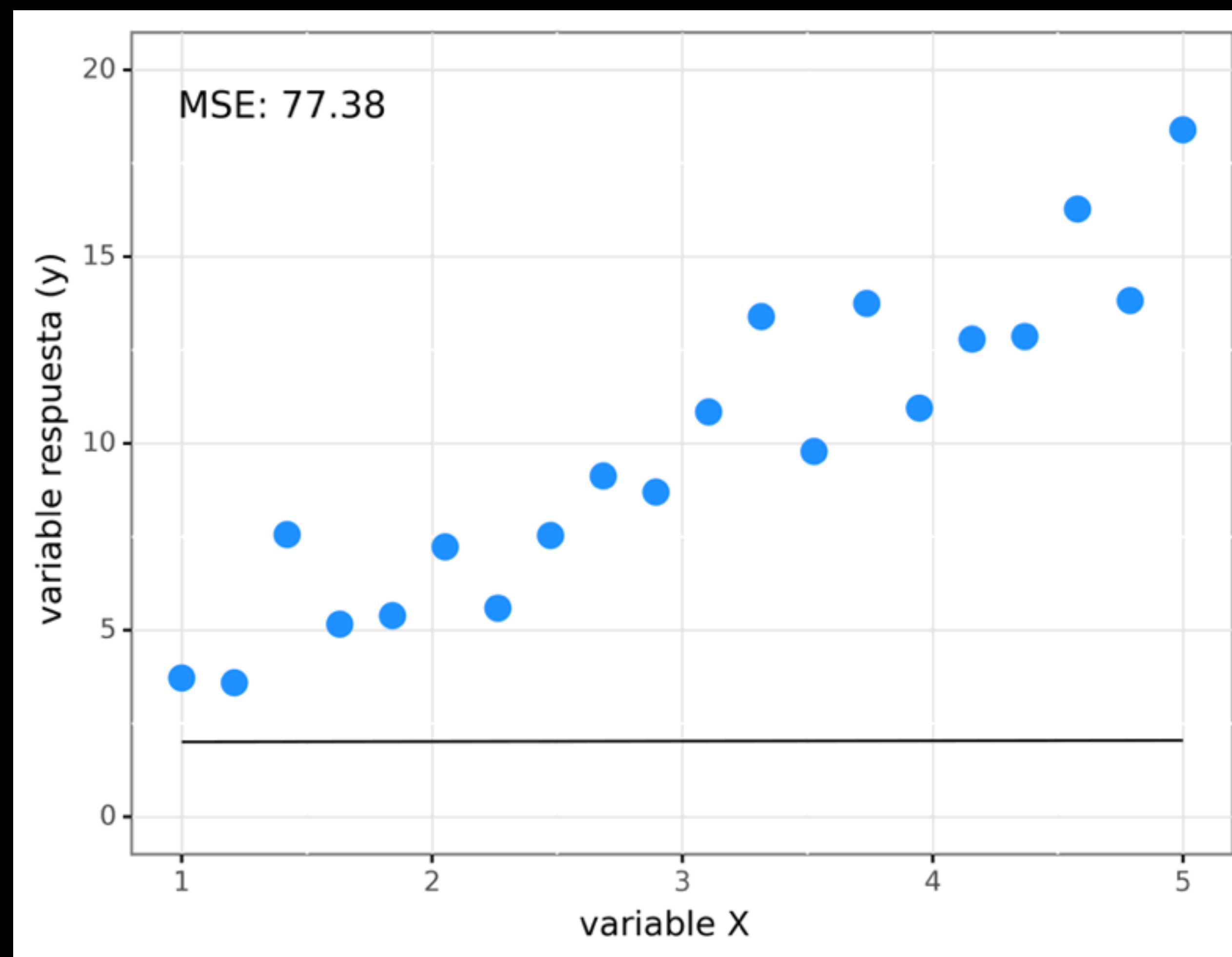
Imagen generada con Sora - OpenAI

Aprende patrones desde los datos



ABC de modelos predictivos

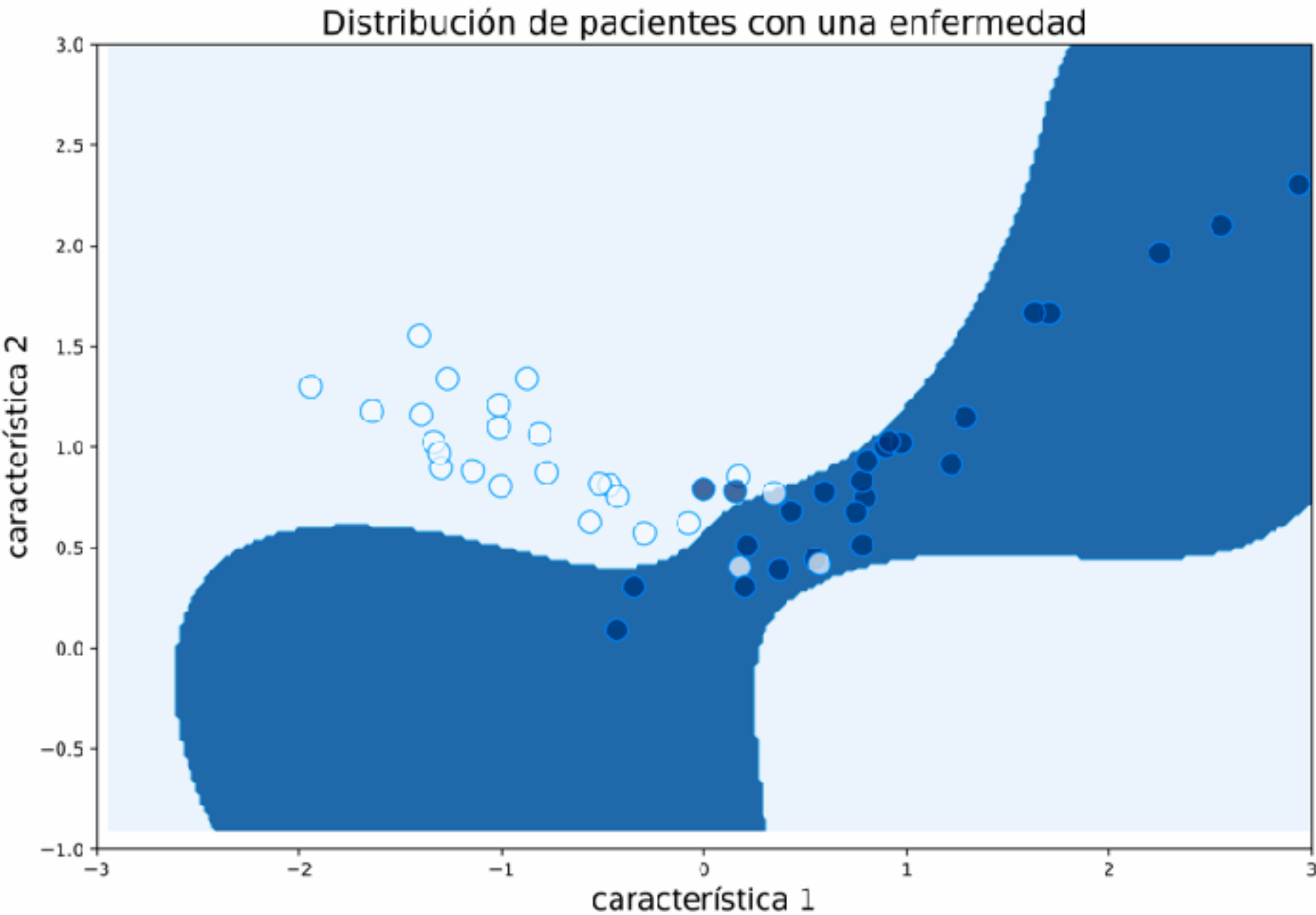
$$f(x) = y \quad \longrightarrow \quad \hat{y} = \beta_0 + \beta_1 X + \epsilon$$



Iteración	MSE
1	77,38
2	
3	
4	
5	
6	
7	
8	
9	
10	

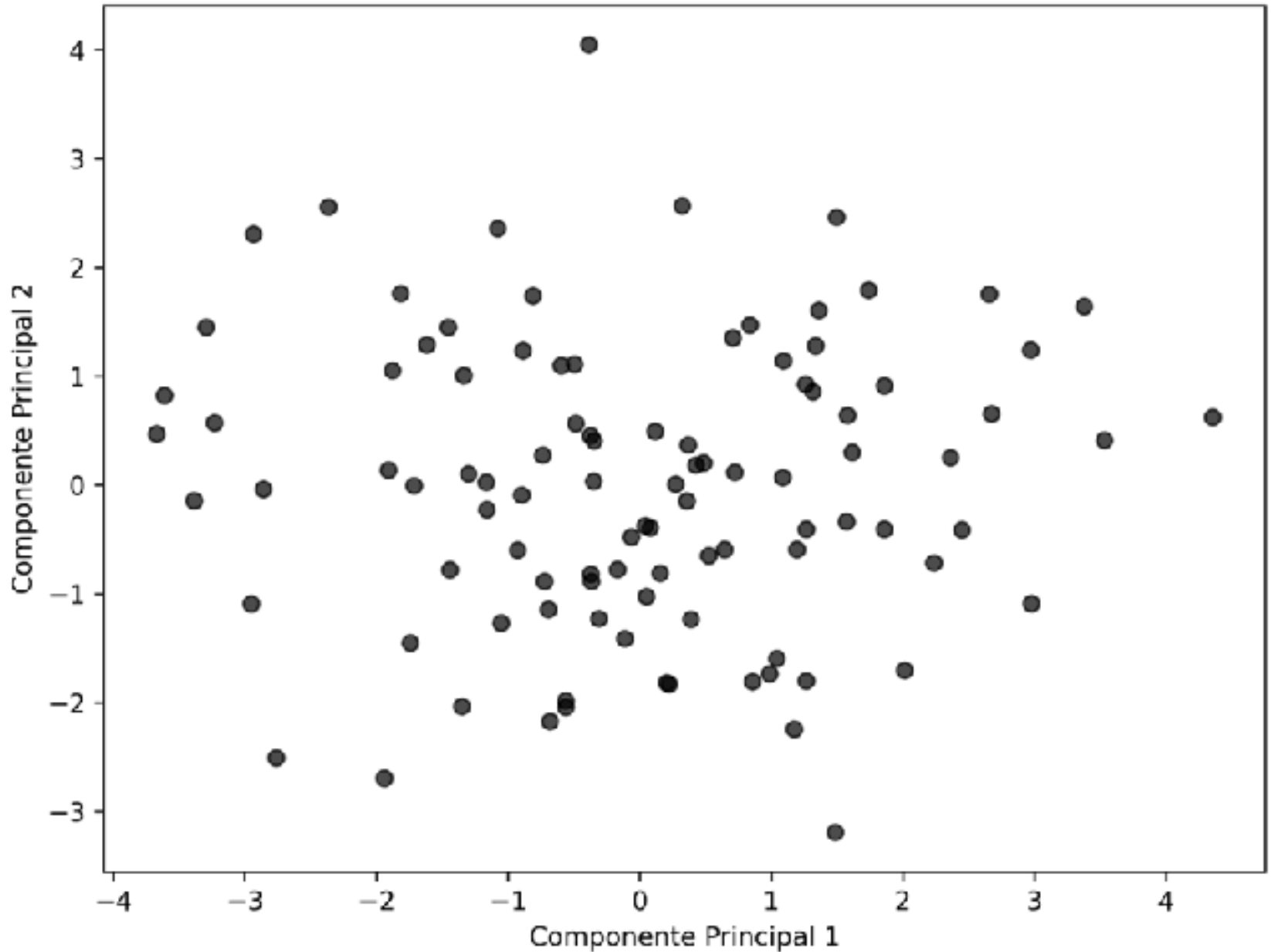
Estrategias

Aprendizaje supervisado



Paciente	Variable 1	Variable 2	Etiqueta
1	0.25	1.6	1
2	0.8	1.3	0
...	...	...	...
N	0.98	0.1	1

Aprendizaje no supervisado



Paciente	Variable 1	Variable 2	Variable 3	...	Variable m
1	0.25	1.6	1.2	...	0.02
2	0.8	1.3	0.1	...	0.30
...	...	...	...	...	...
n	0.98	0.1	2.4	...	0.12

No hay
etiqueta

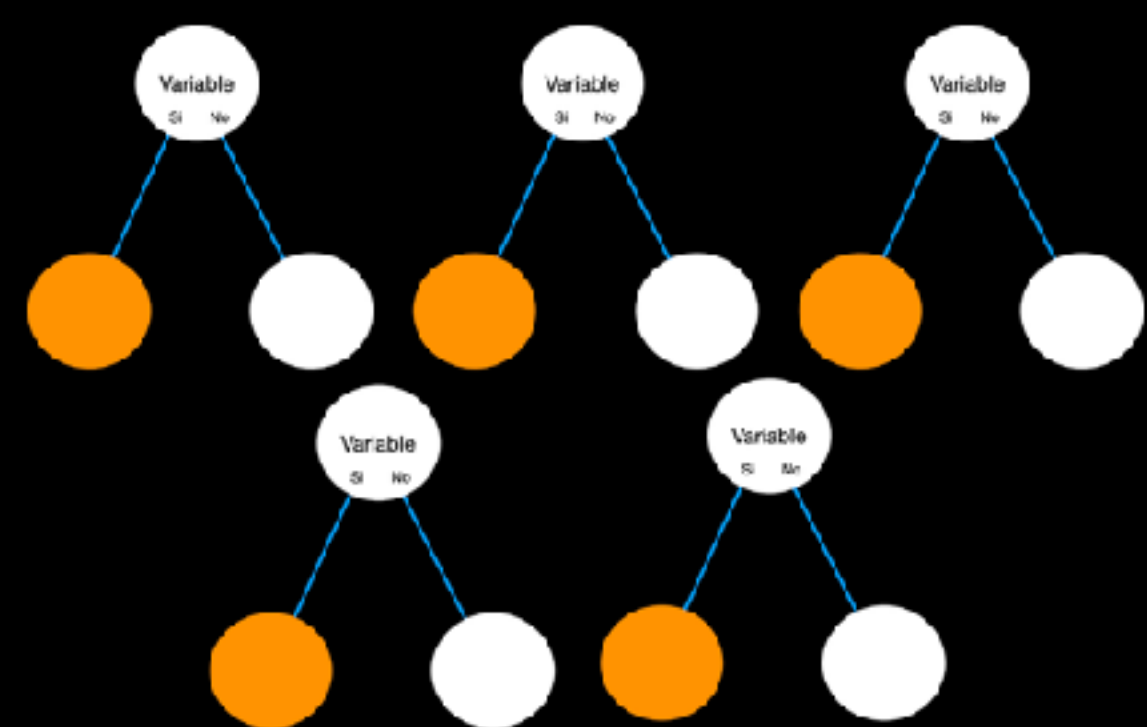
Ejemplos de modelos y usos

Estrategia supervisada

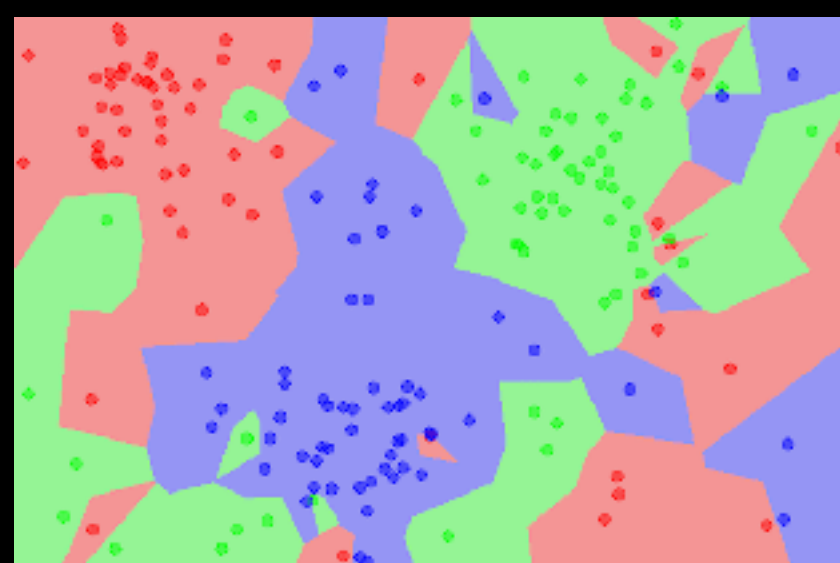
Ensamblados: RF, GB, XGB

Datos tabulares

Redes Neuronales

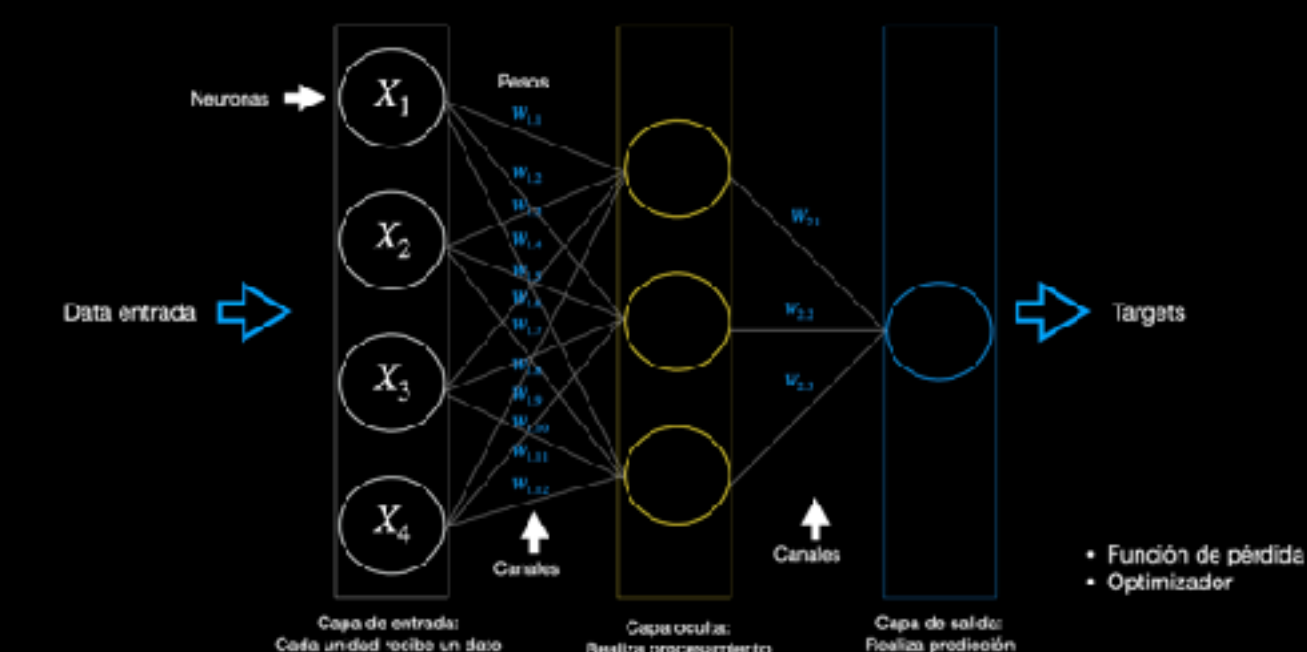
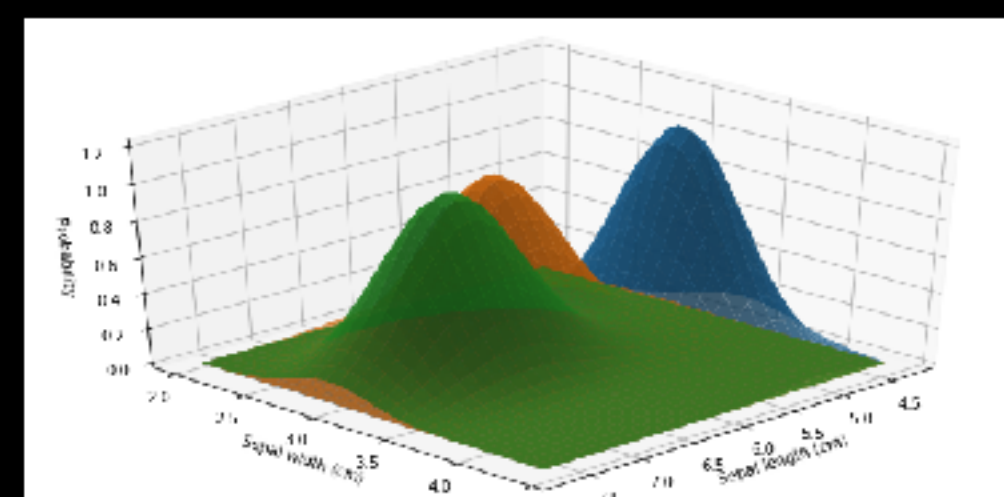


KNN

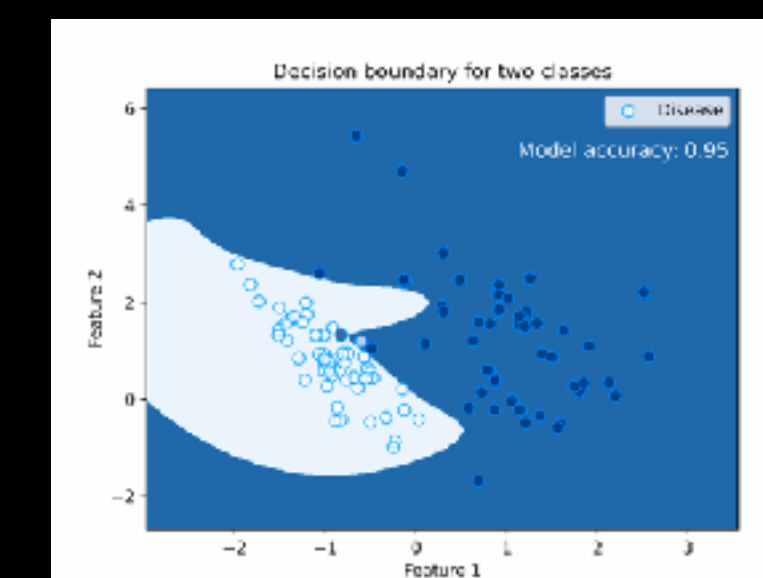


Paciente	Variable 1	Variable 2	Etiqueta
1	0.25	1.6	1
2	0.8	1.3	0
...	...	...	...
N	0.98	0.1	1

Naive Bayes



SVM



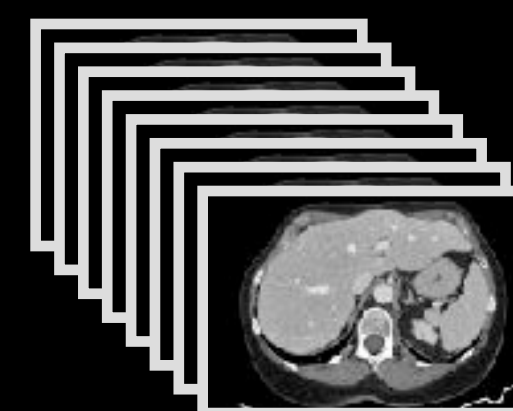
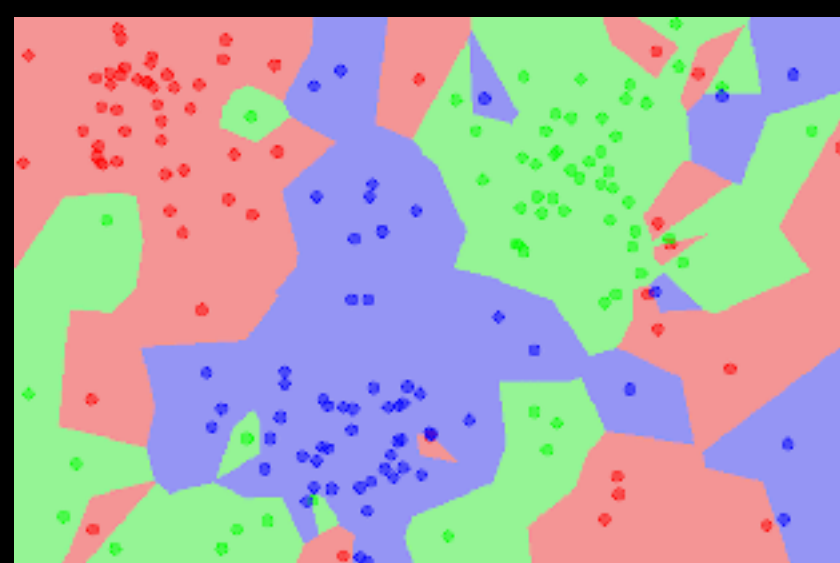
Ejemplos de modelos y usos

Estrategia supervisada

Árboles de decisión

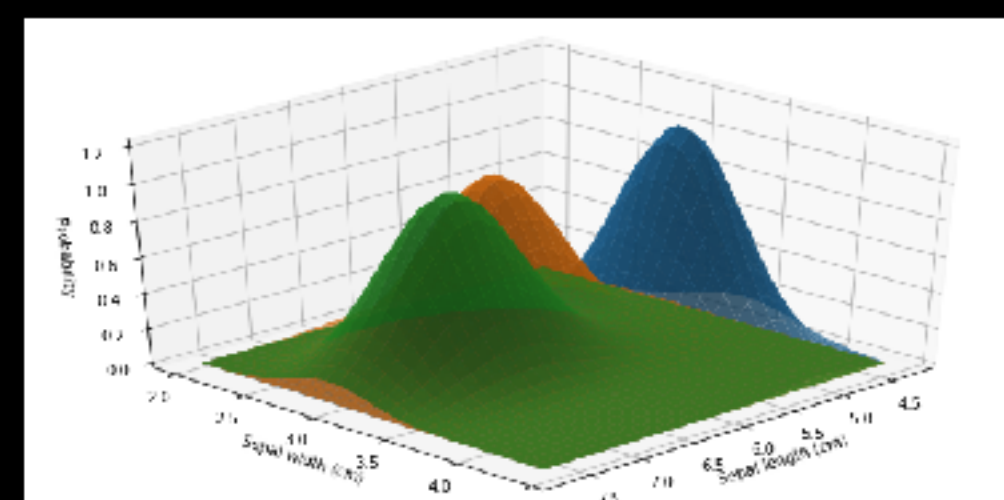


KNN

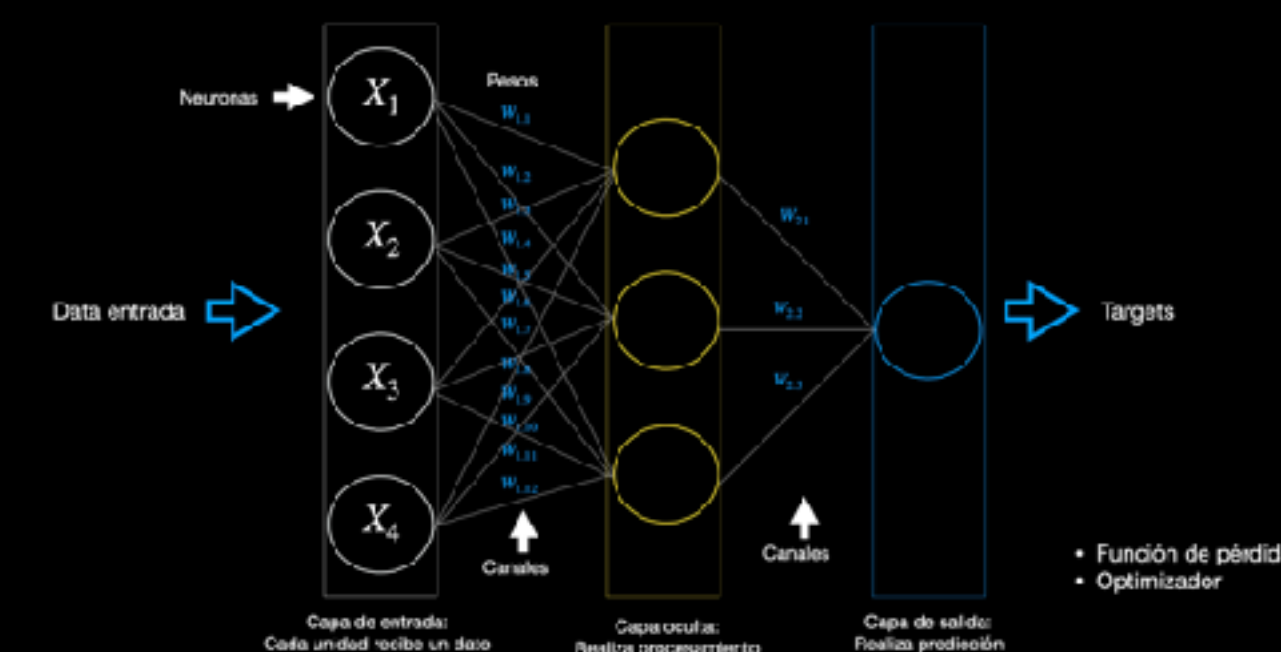


Paciente	Variable 1	Variable 2	Etiqueta
1	0.25	1.6	1
2	0.8	1.3	0
...	...	...	...
N	0.98	0.1	1

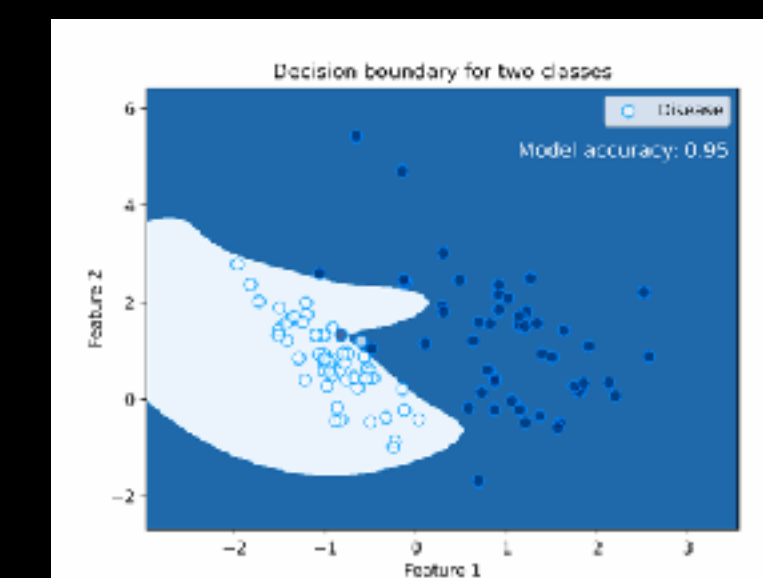
Naive Bayes



Redes Neuronales

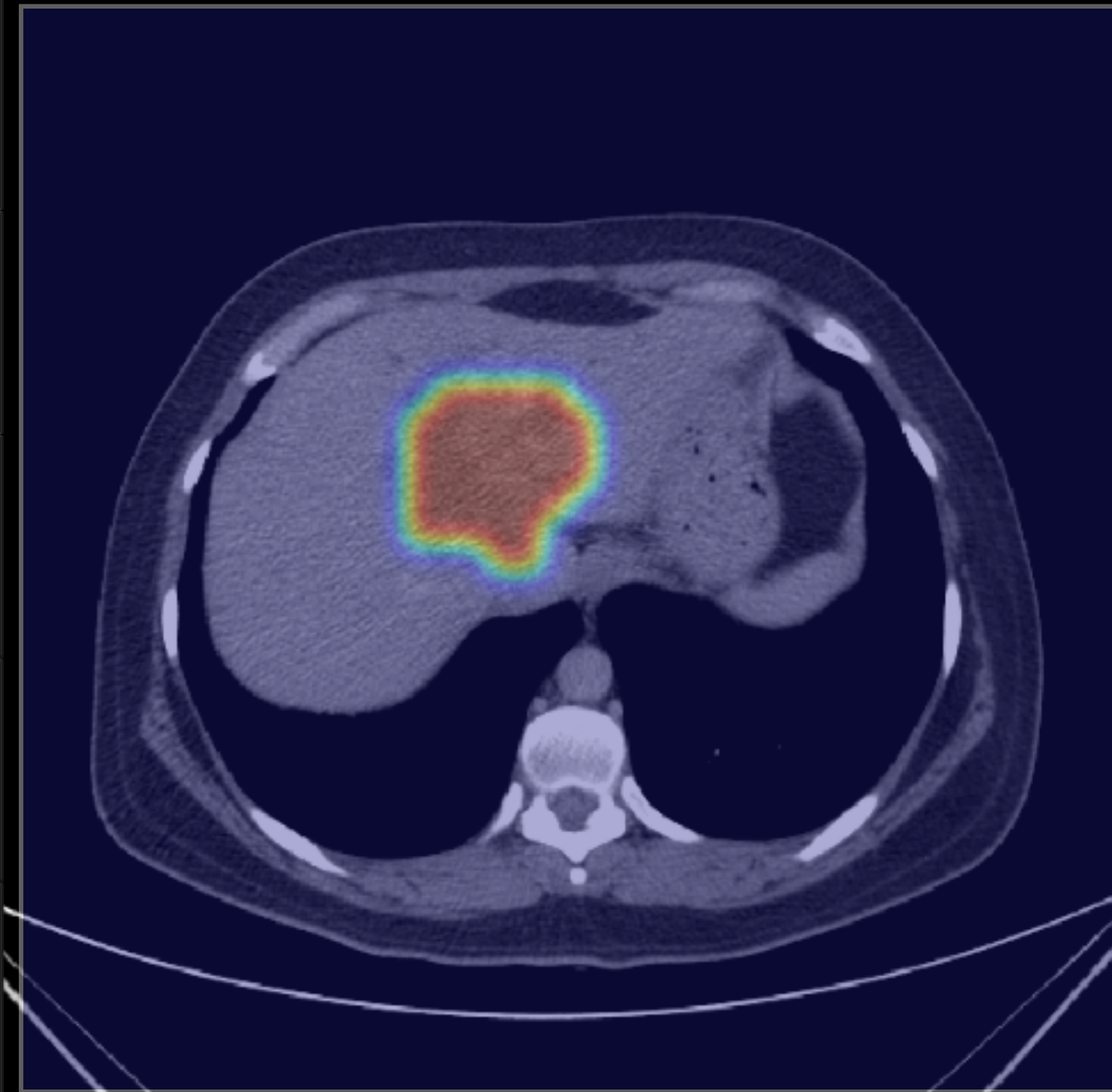


SVM



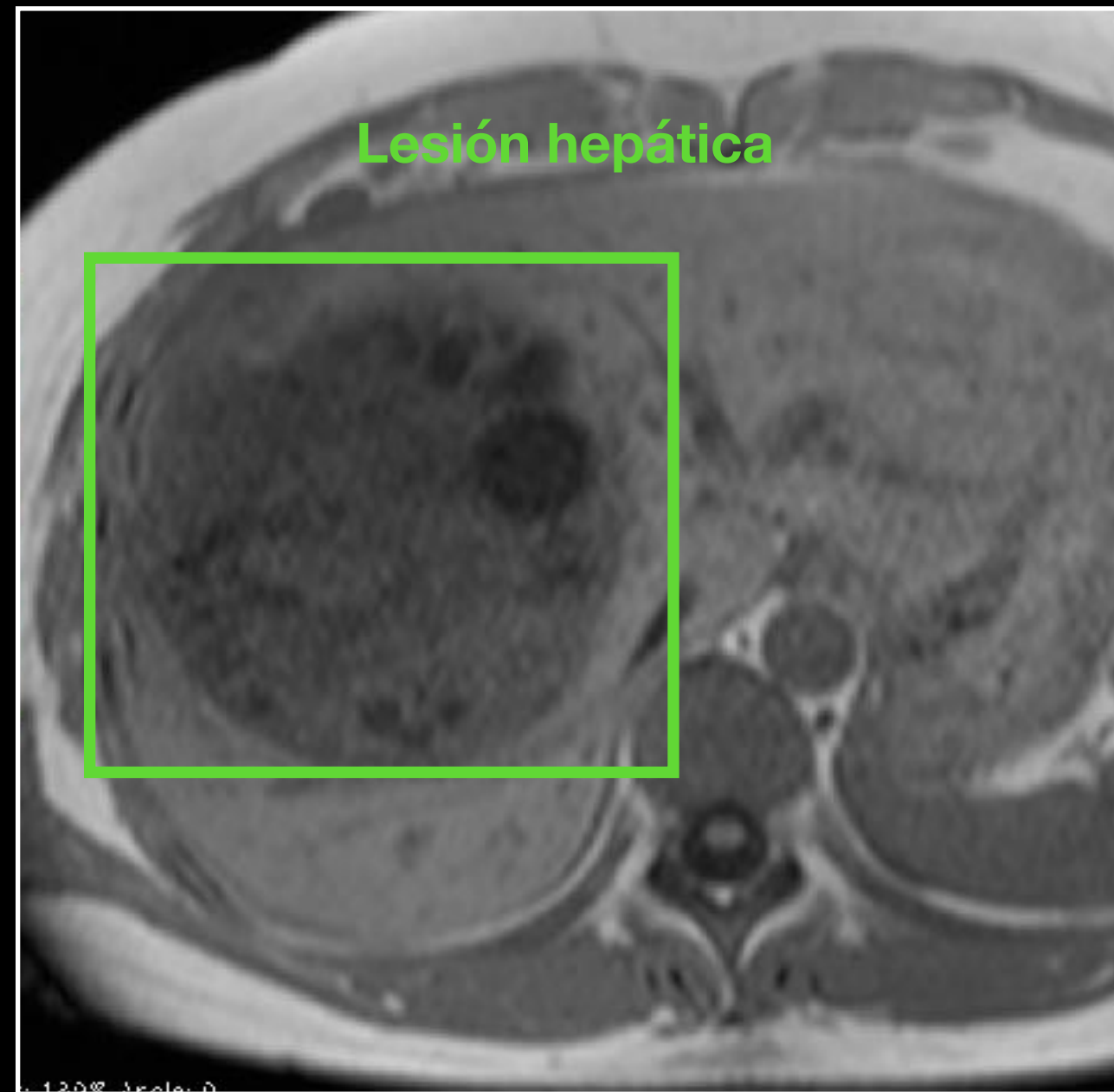
Ejemplos de modelos y usos

Estrategia supervisada: *Deep Learning*

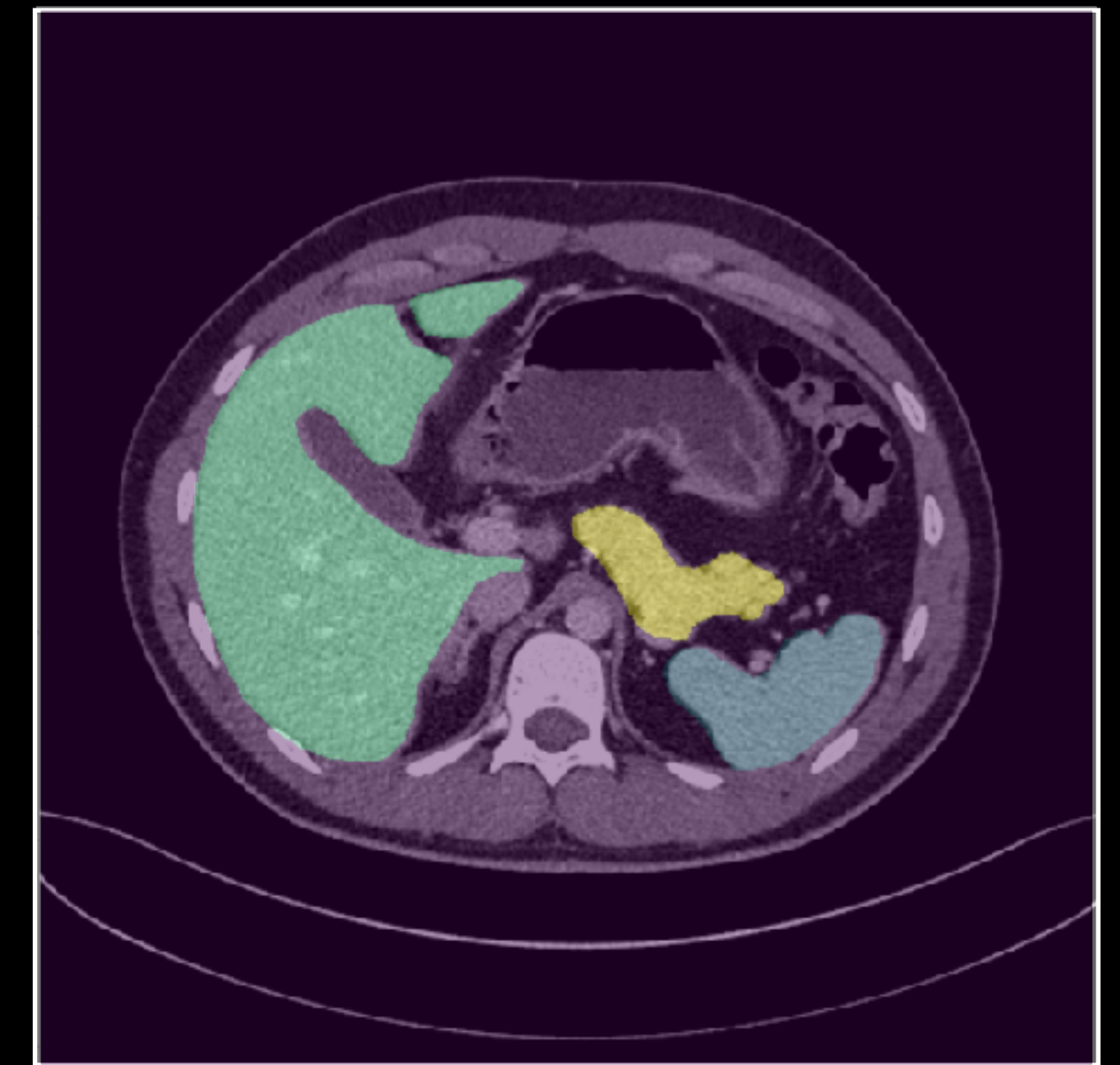


Clasificación

Etiqueta



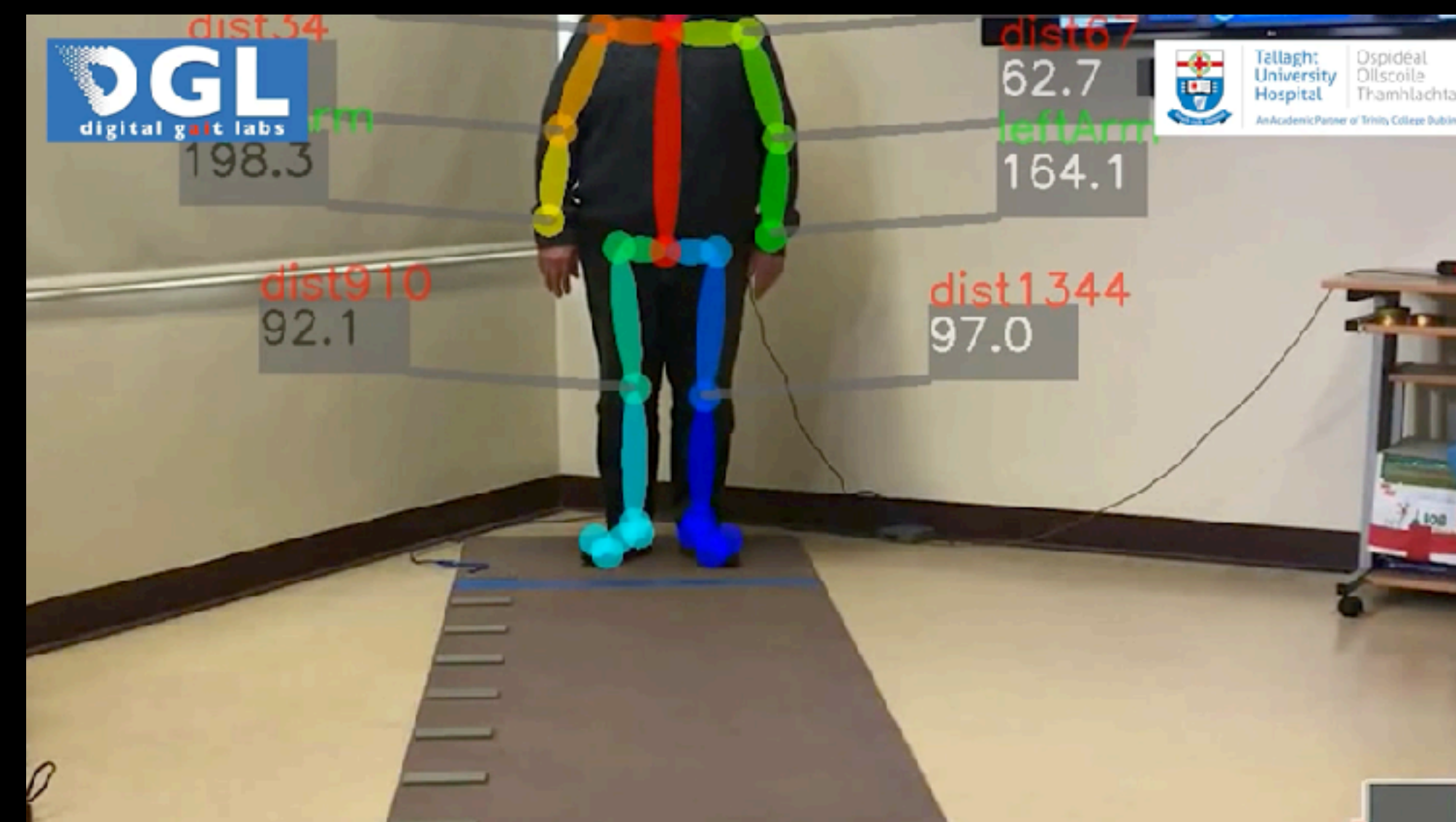
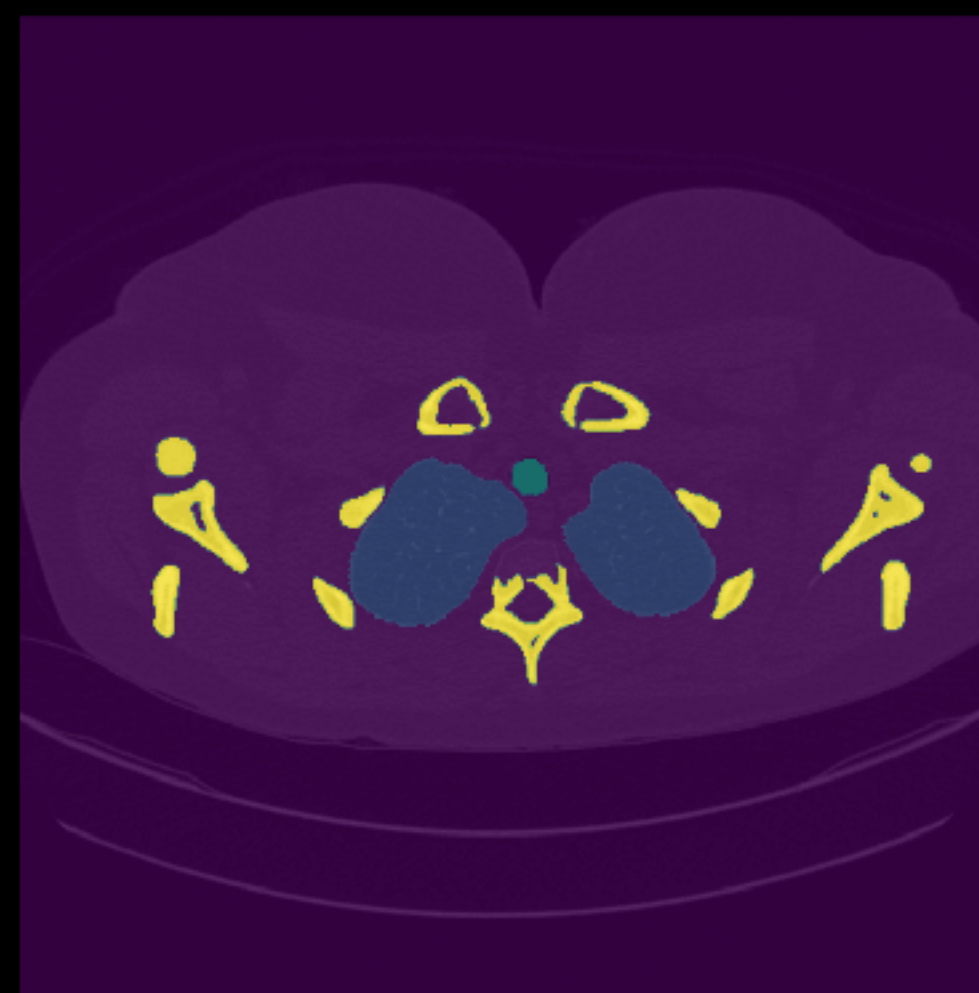
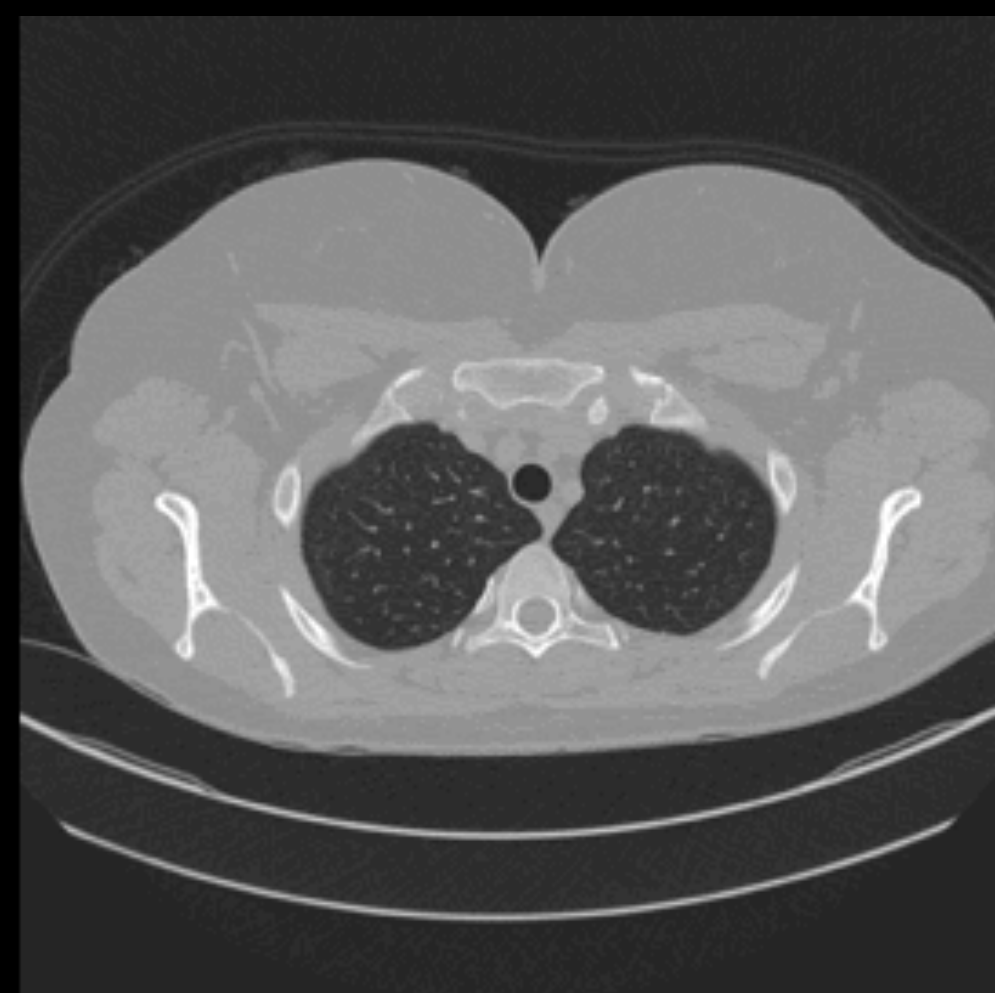
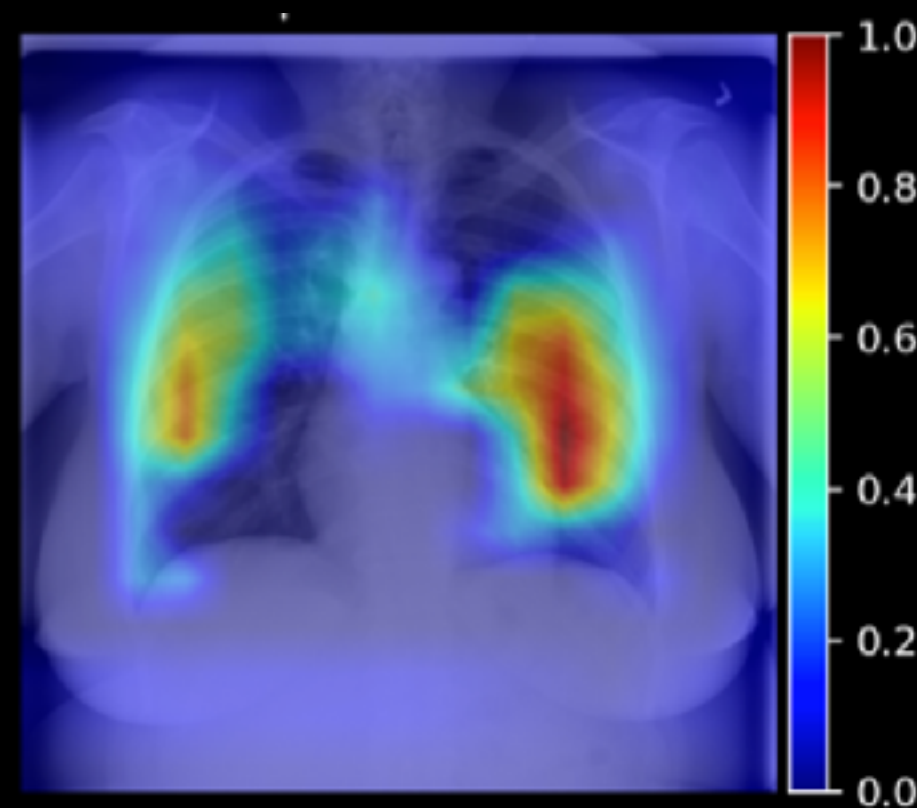
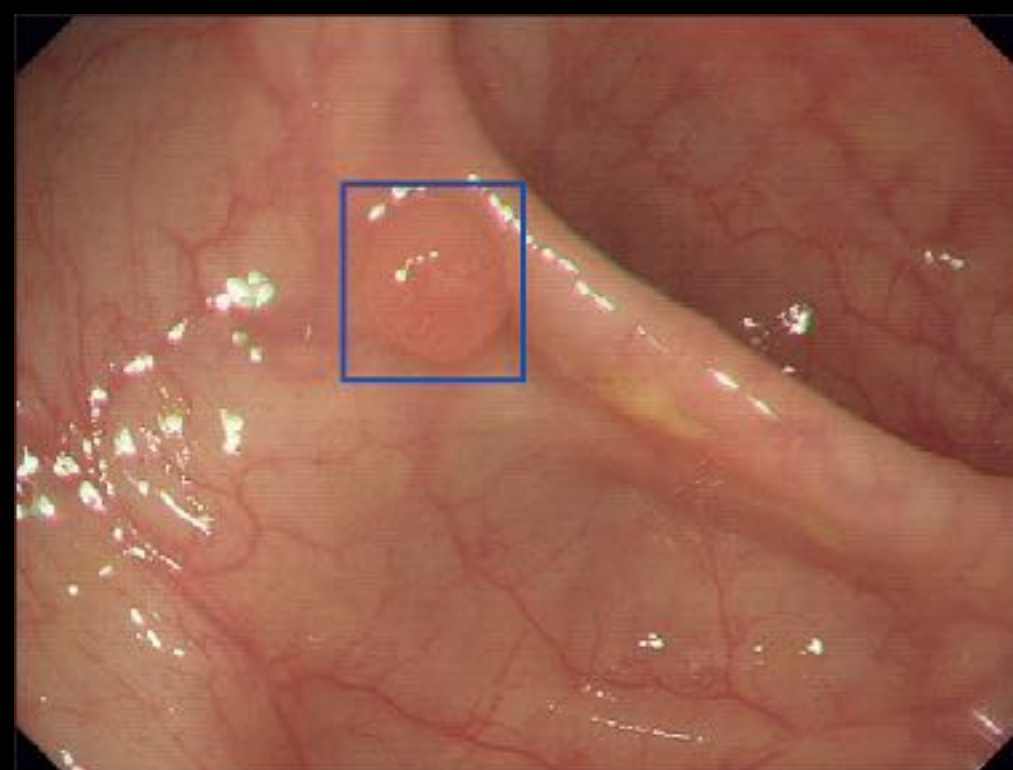
Detección



Segmentación semántica

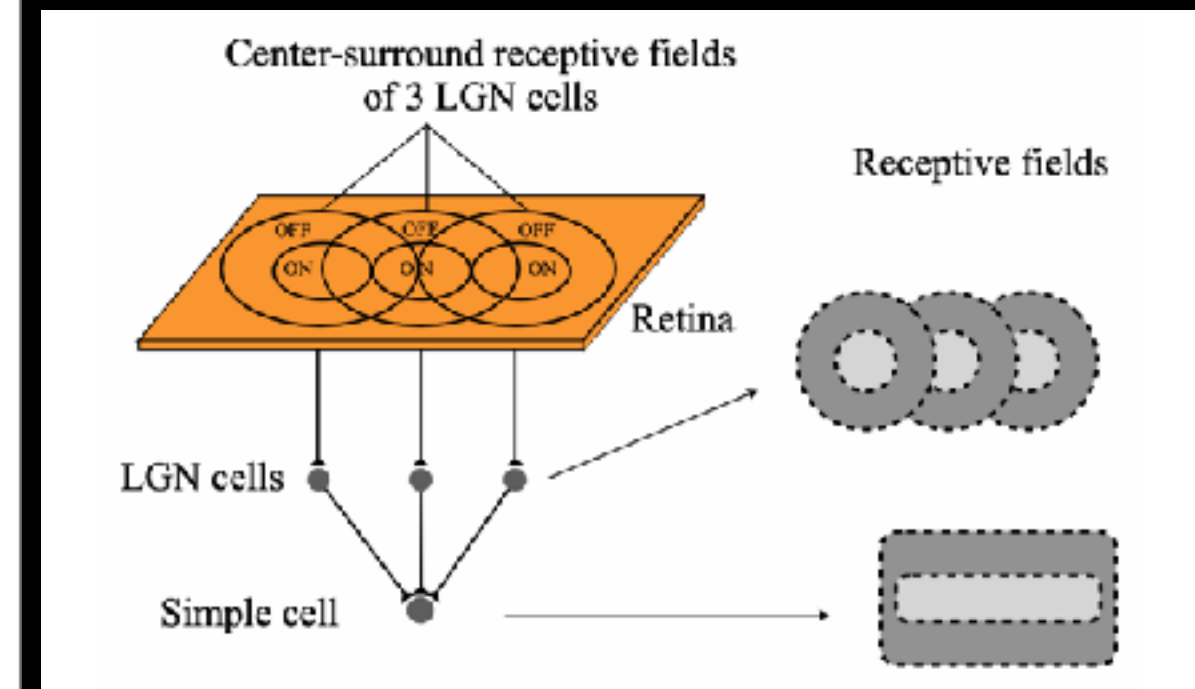
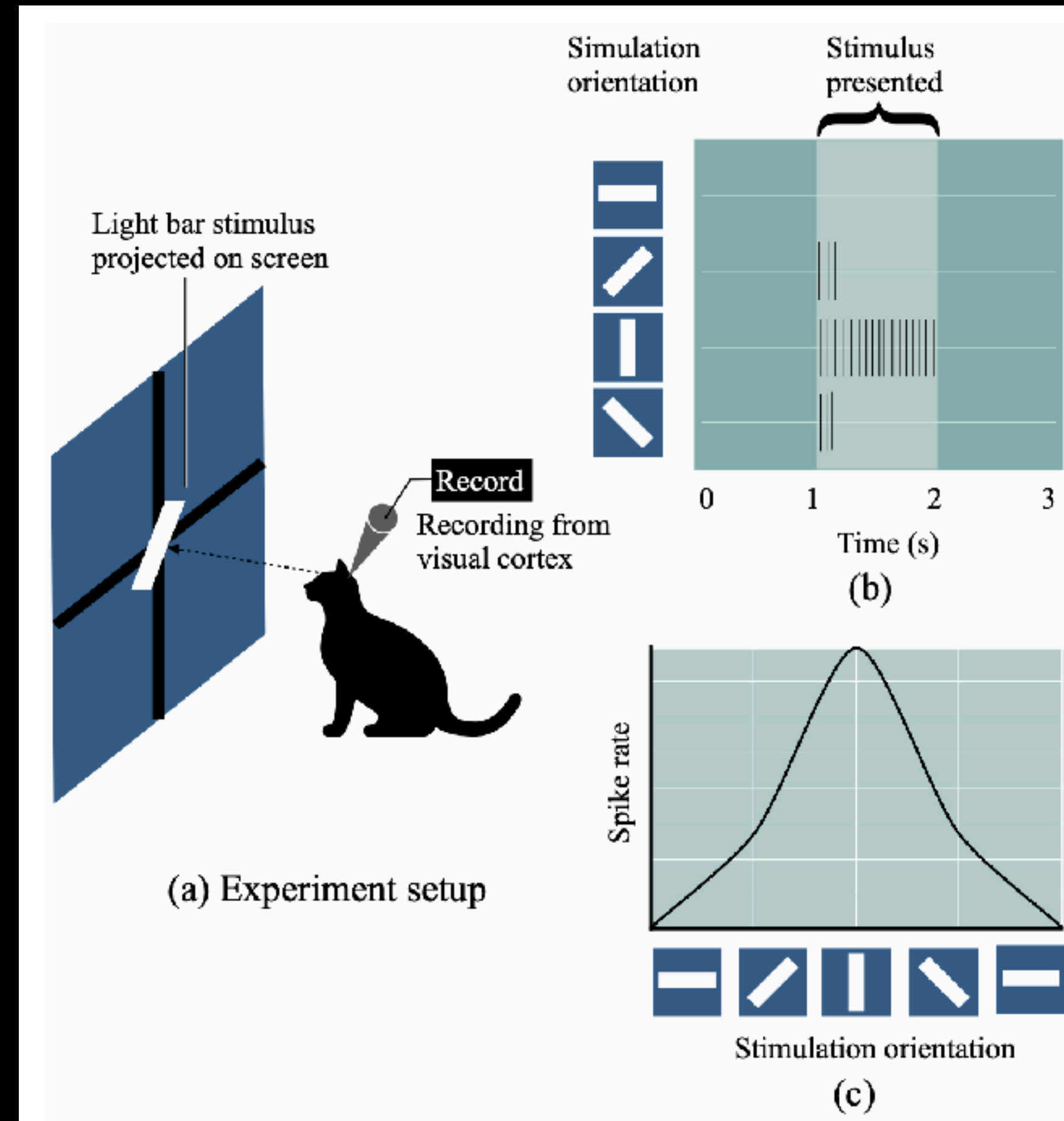
Anotación

El mundo es visión



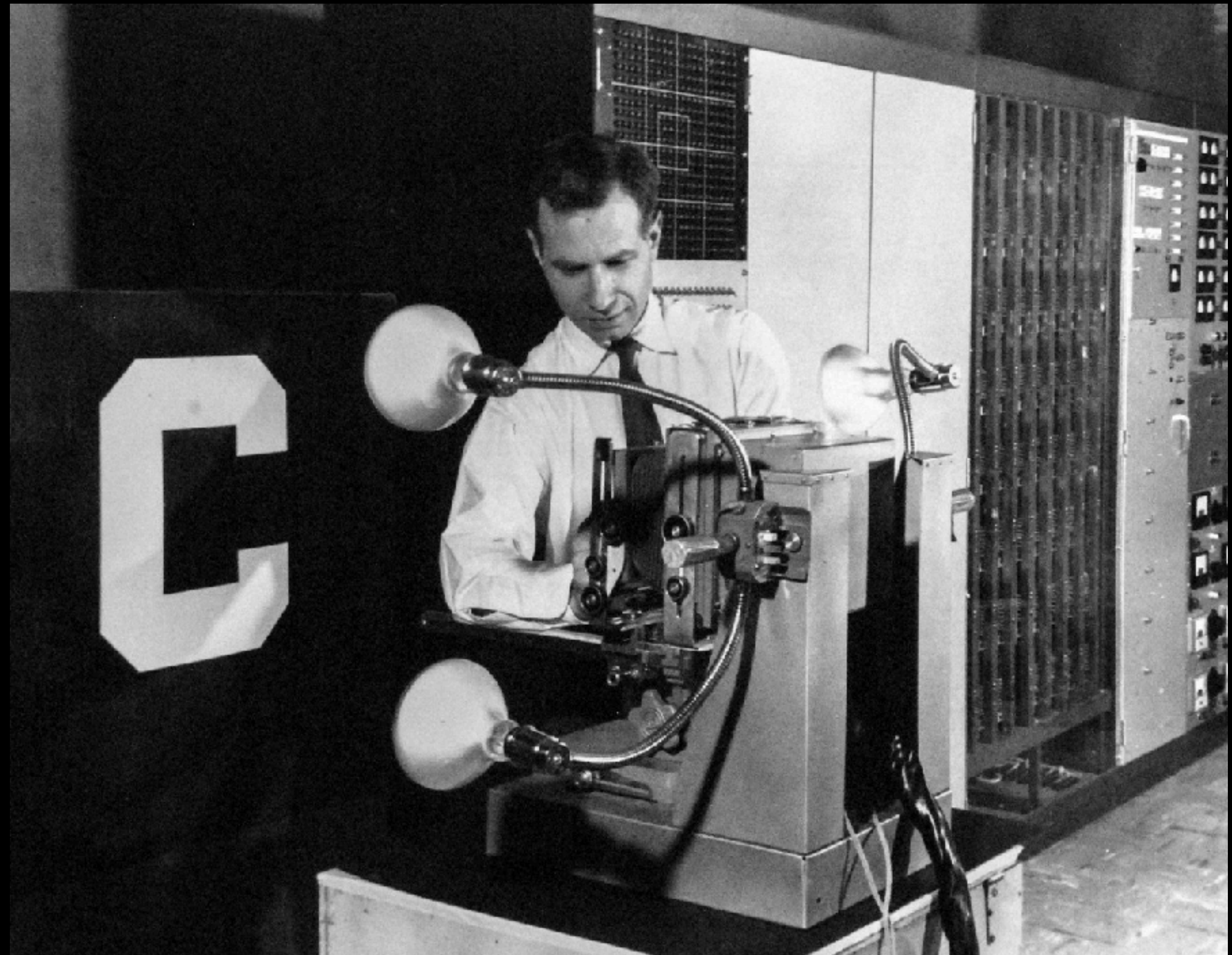
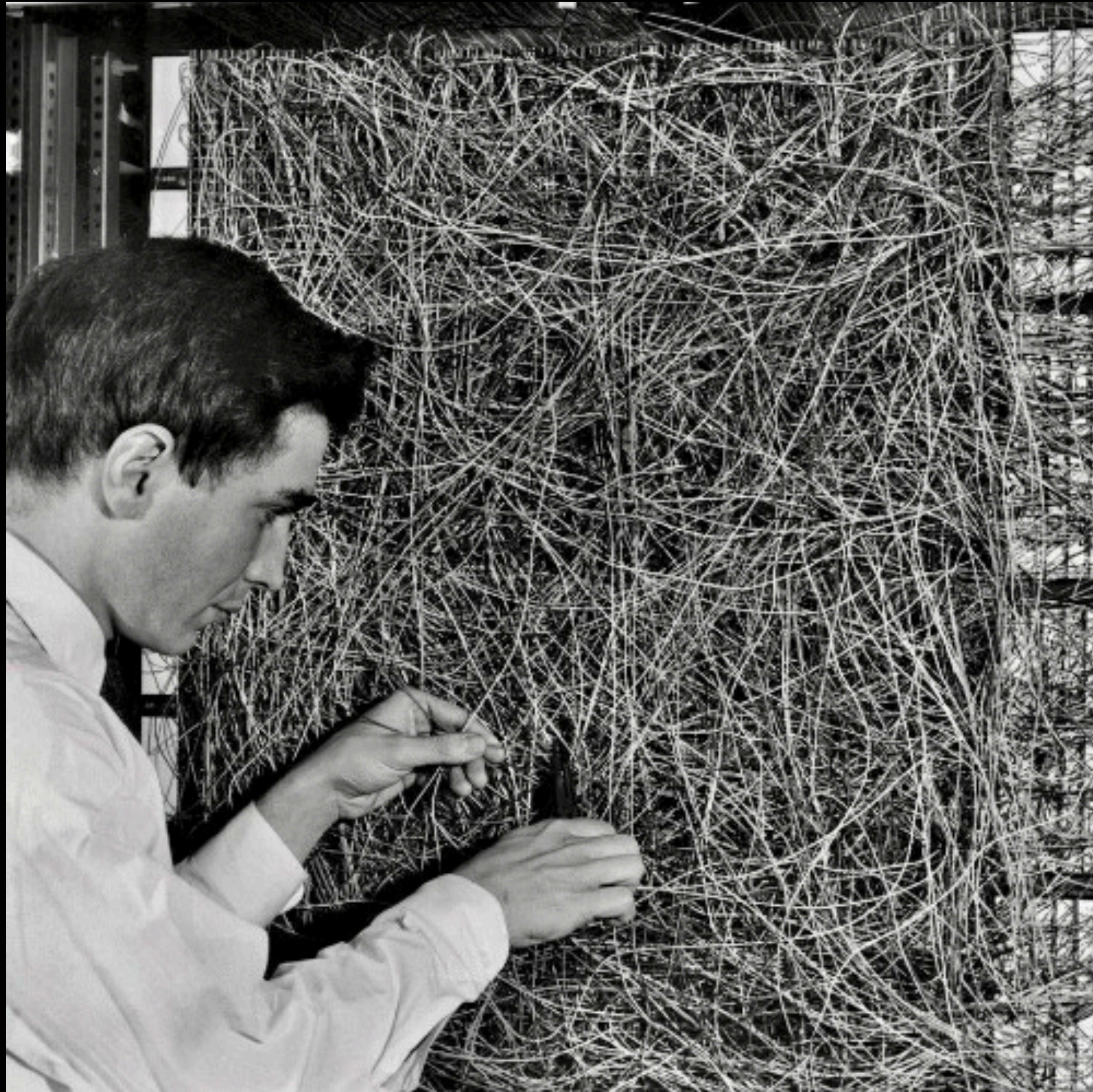
Introducción

- McCulloch-Pitts (1943):
 - Modelo matemático de neurona.
- Frank Rosenblatt (1958):
 - Perceptrón
- Hubel & Wiesel (1961):
 - Describen modelo neurobiológico de selectividad neuronal por orientación.

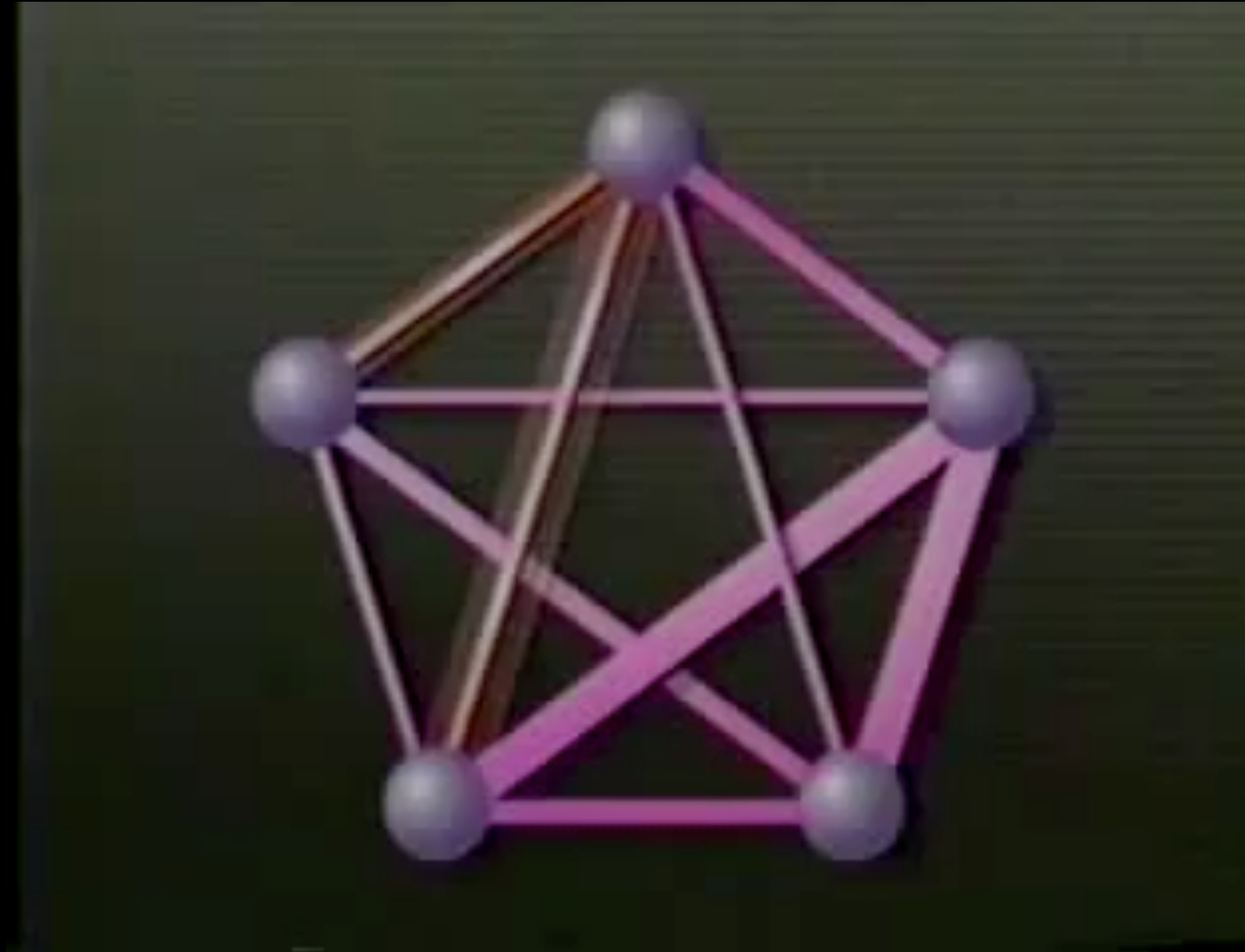


Perceptrón - Mark I: 1958

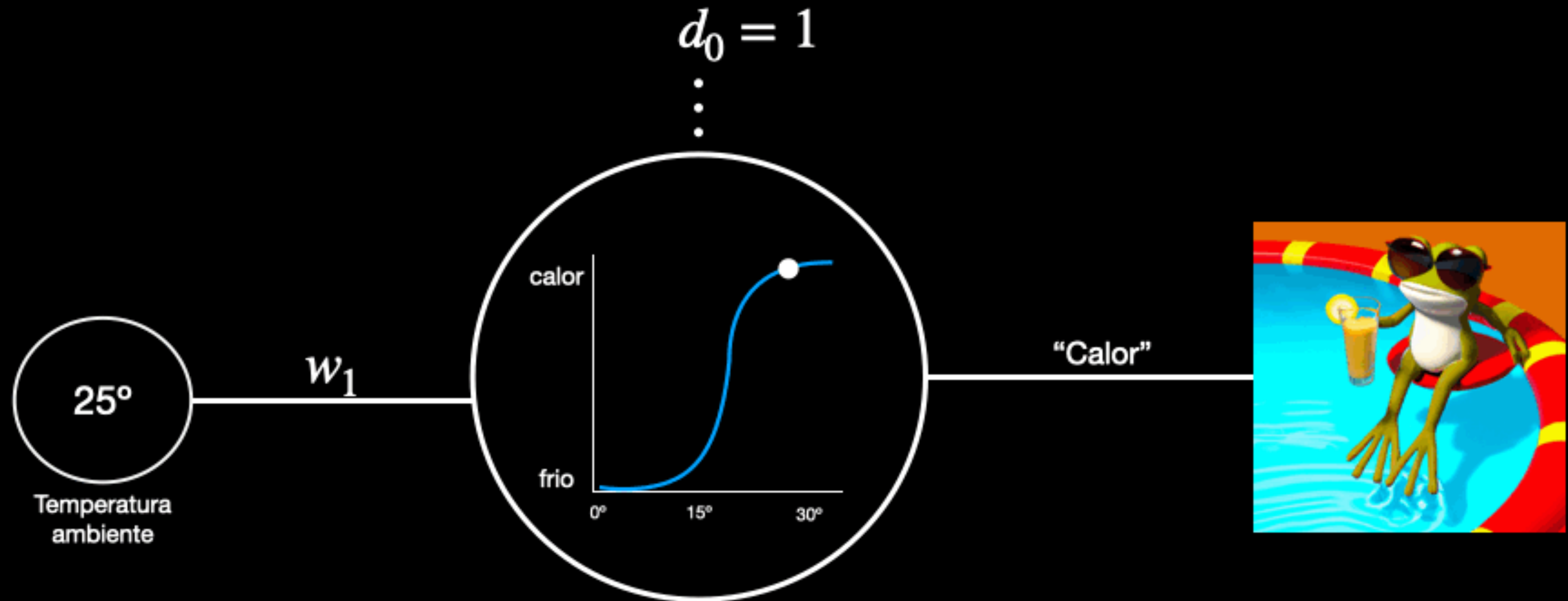
- Frank Rosenblatt



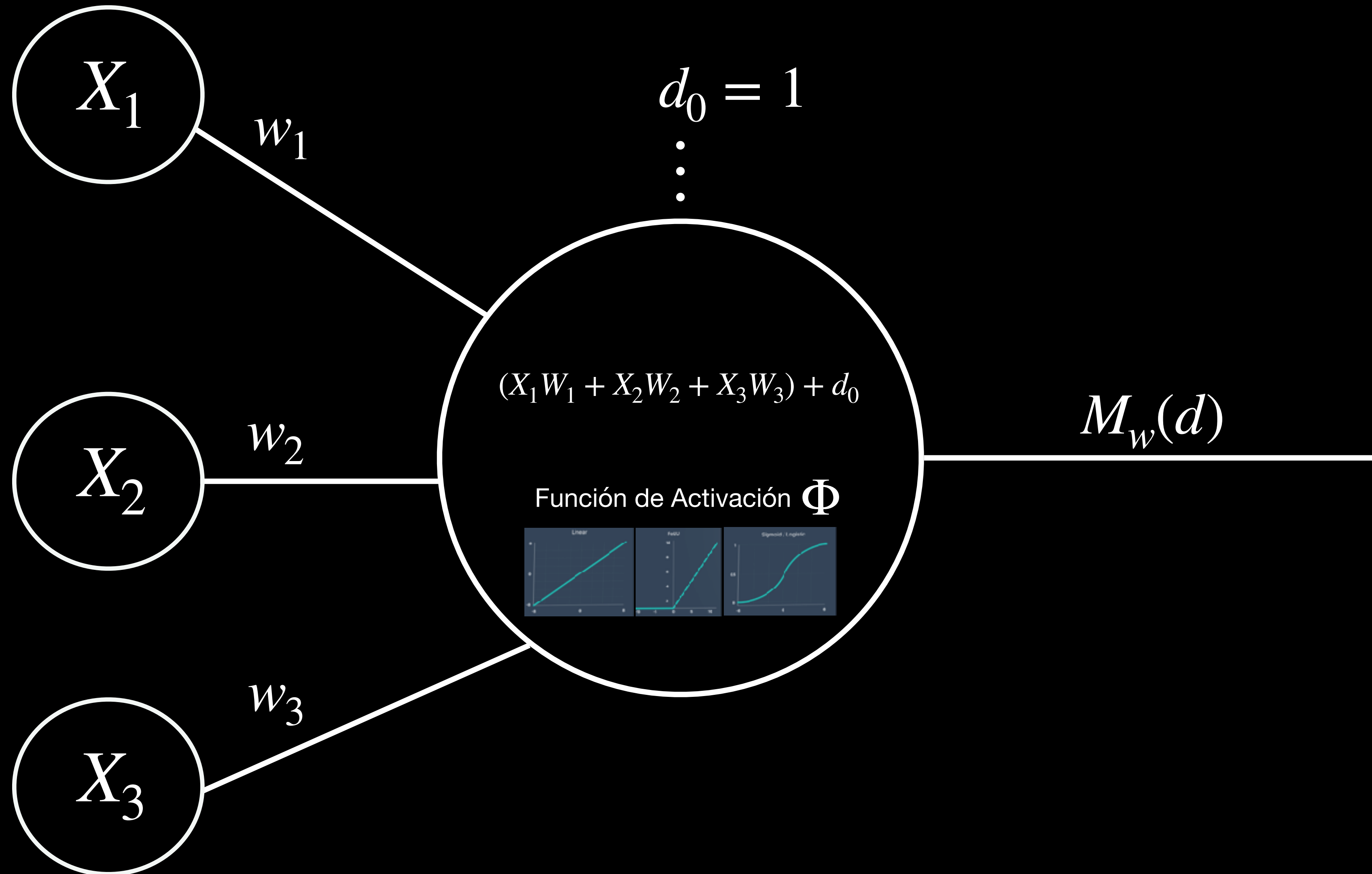
Perceptrón - Mark I: 1958



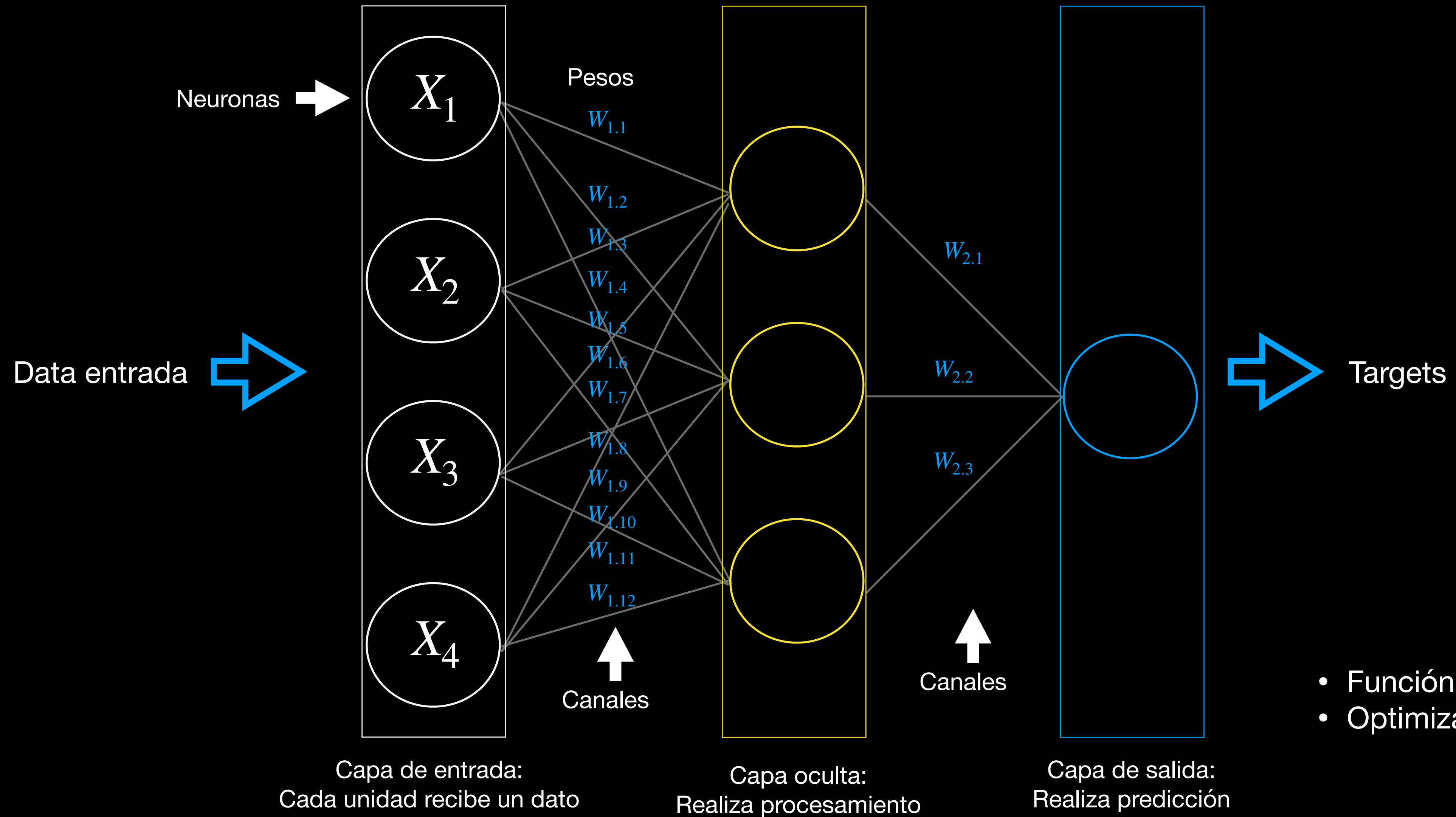
¿Qué es una neurona?



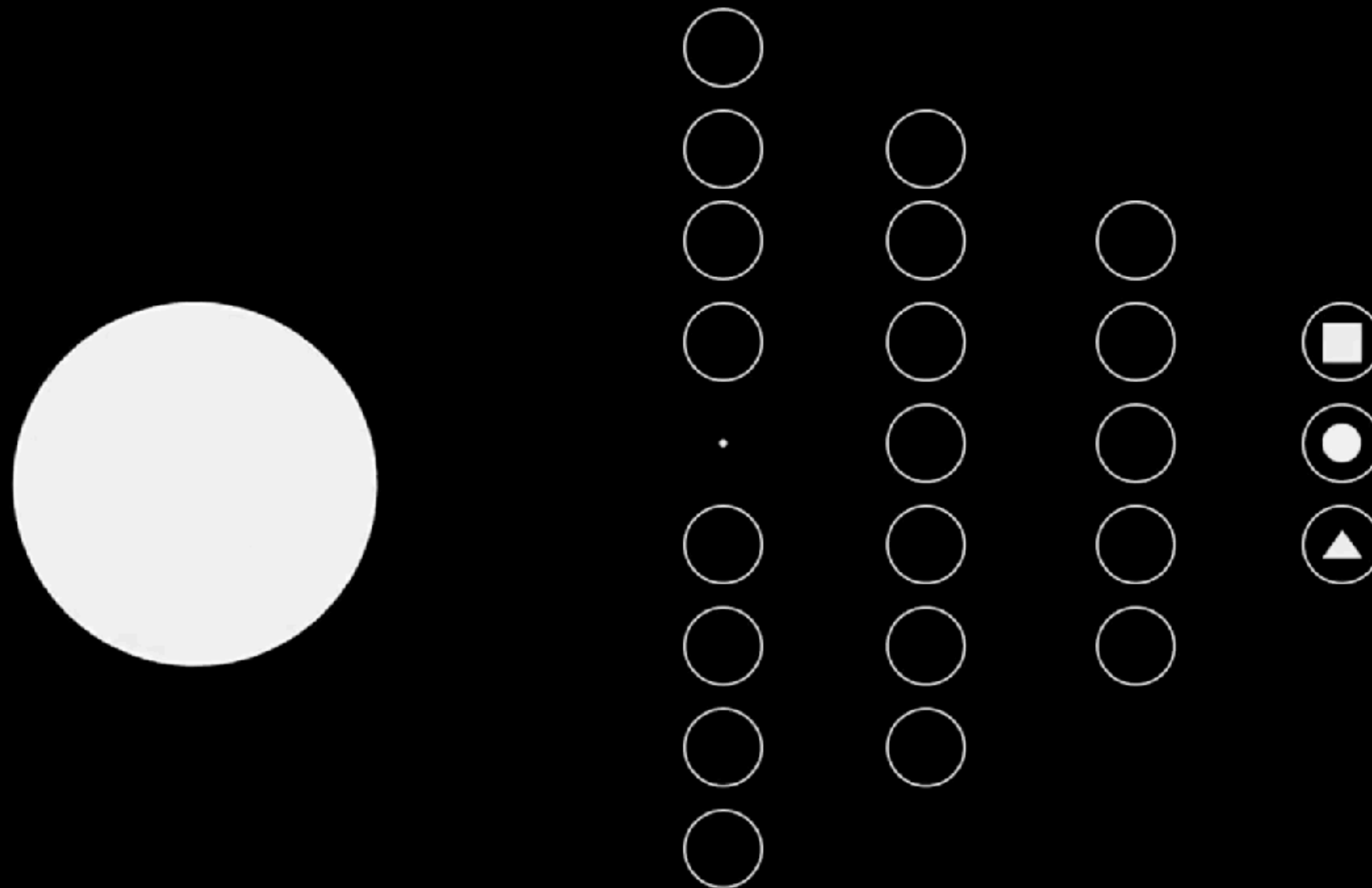
¿Qué es una neurona?



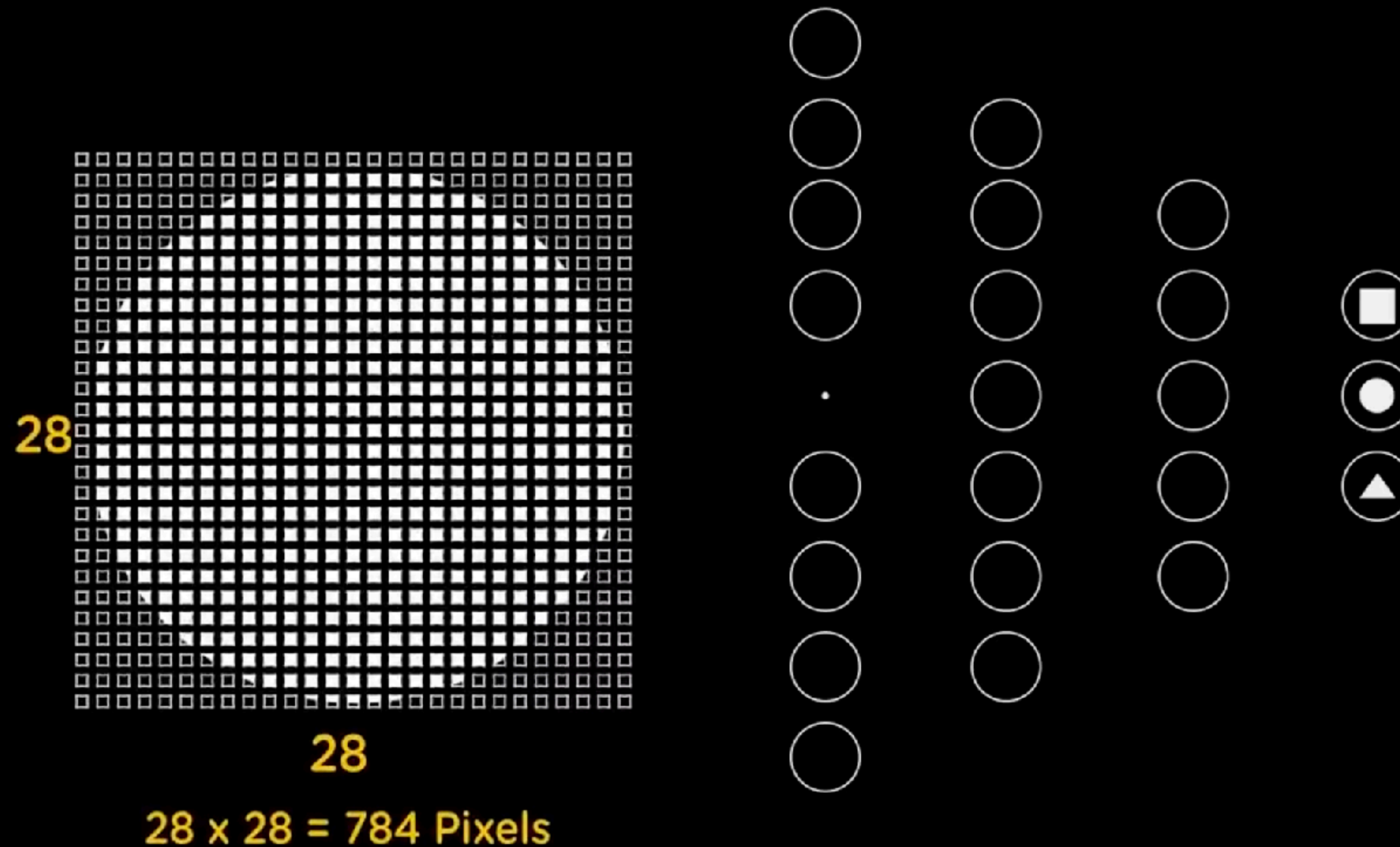
Anatomía de una red neuronal: “feed forward”



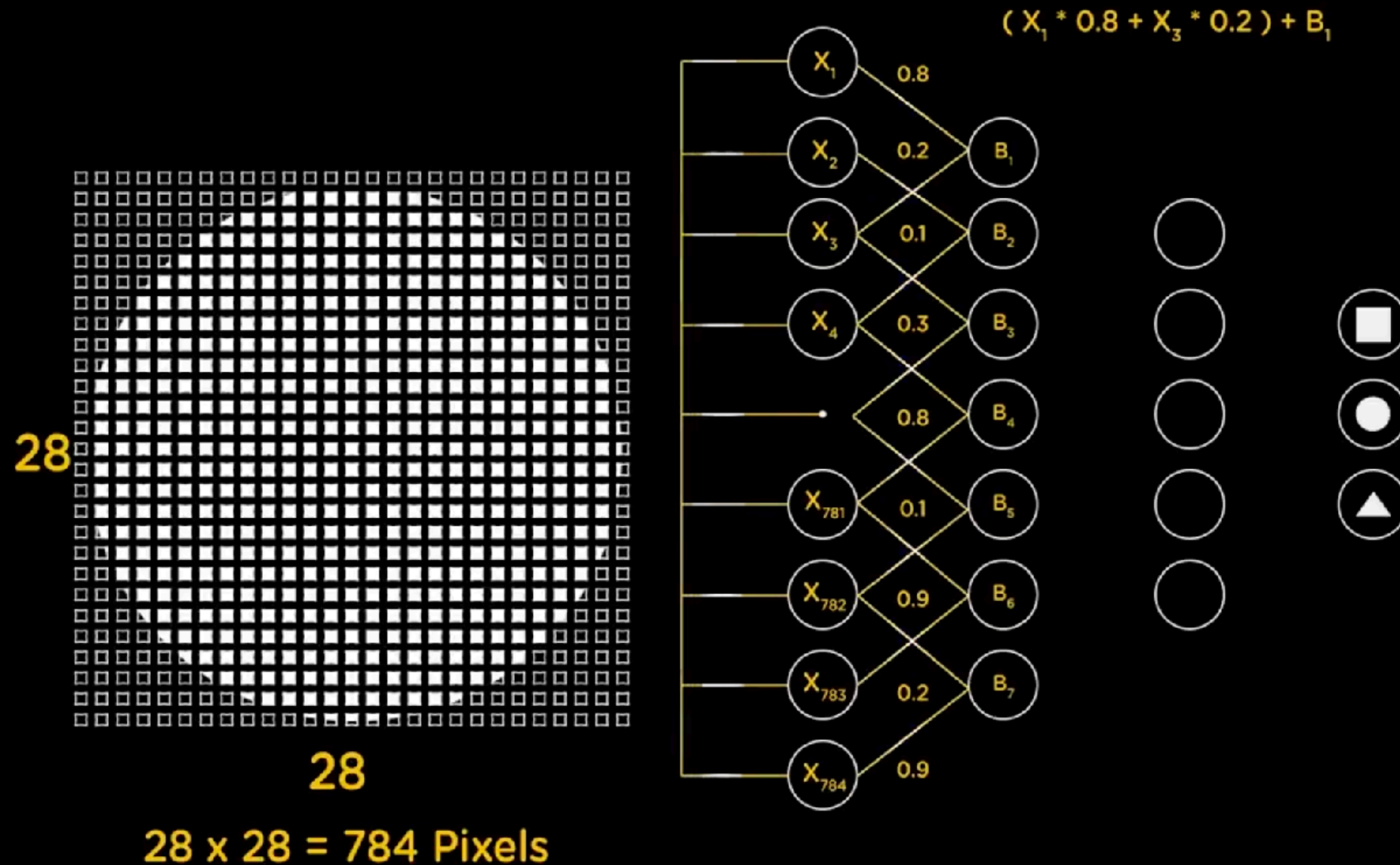
Funcionamiento red “feed forward”



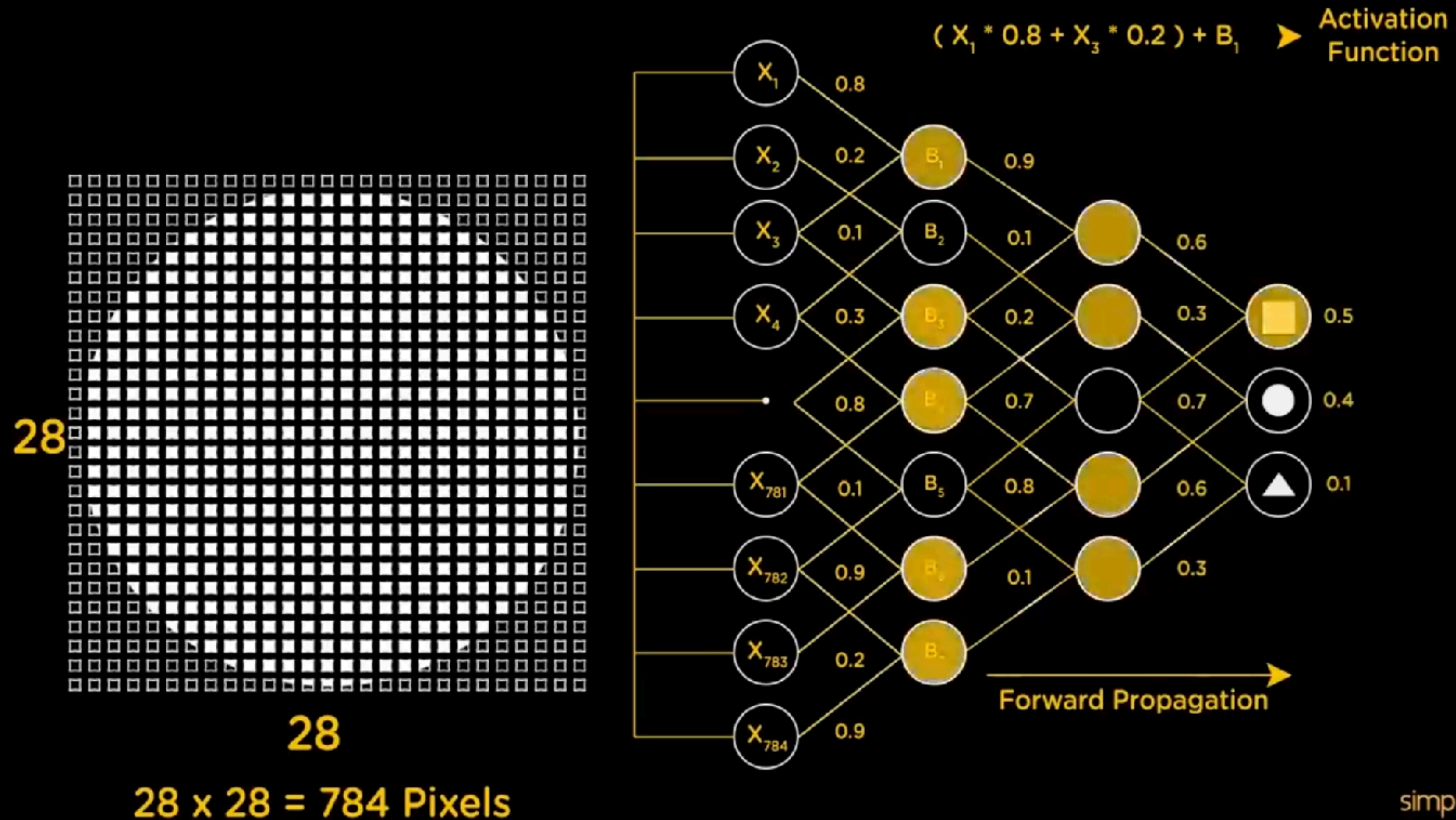
Funcionamiento red “feed forward”



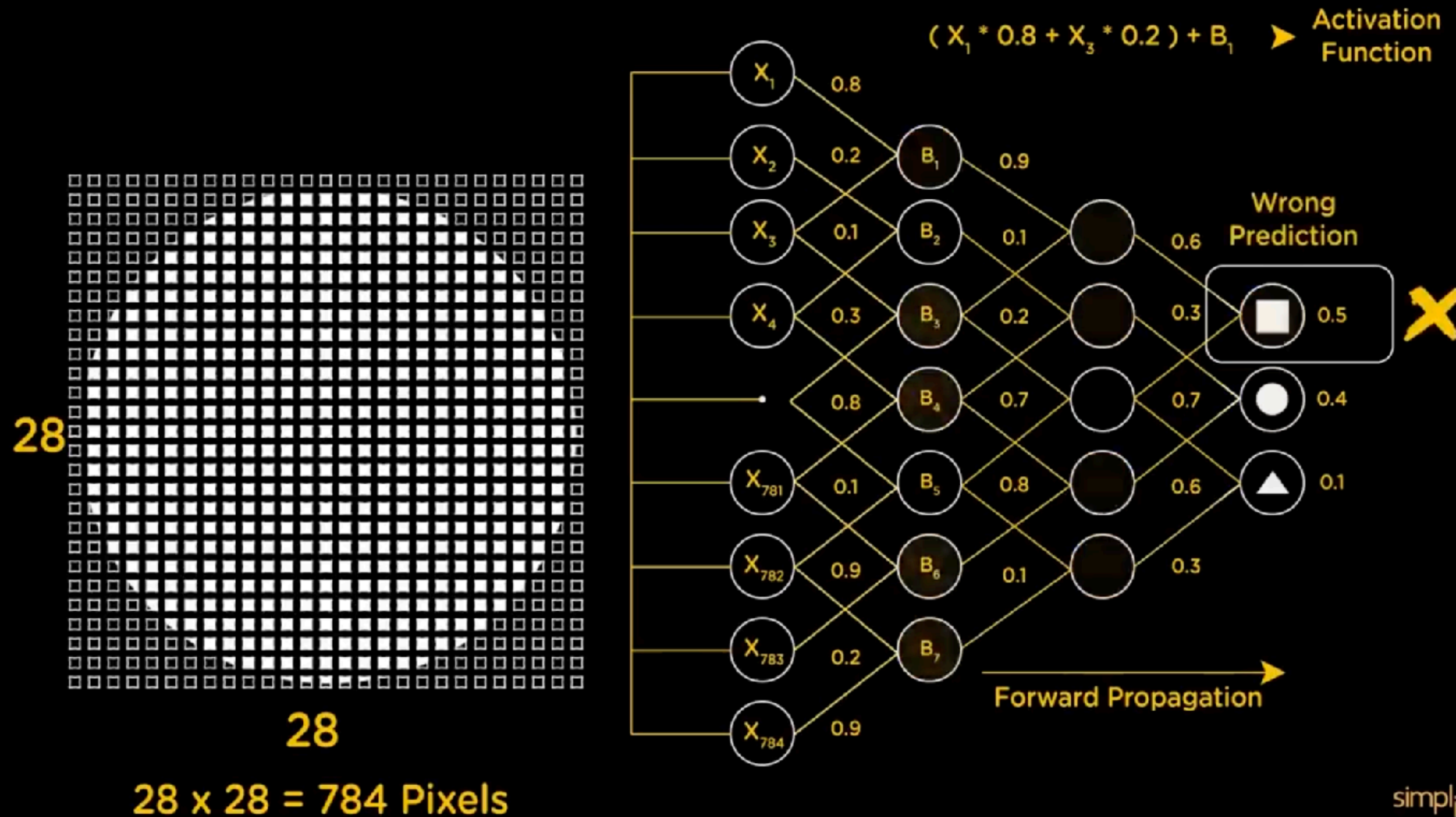
Funcionamiento red “feed forward”



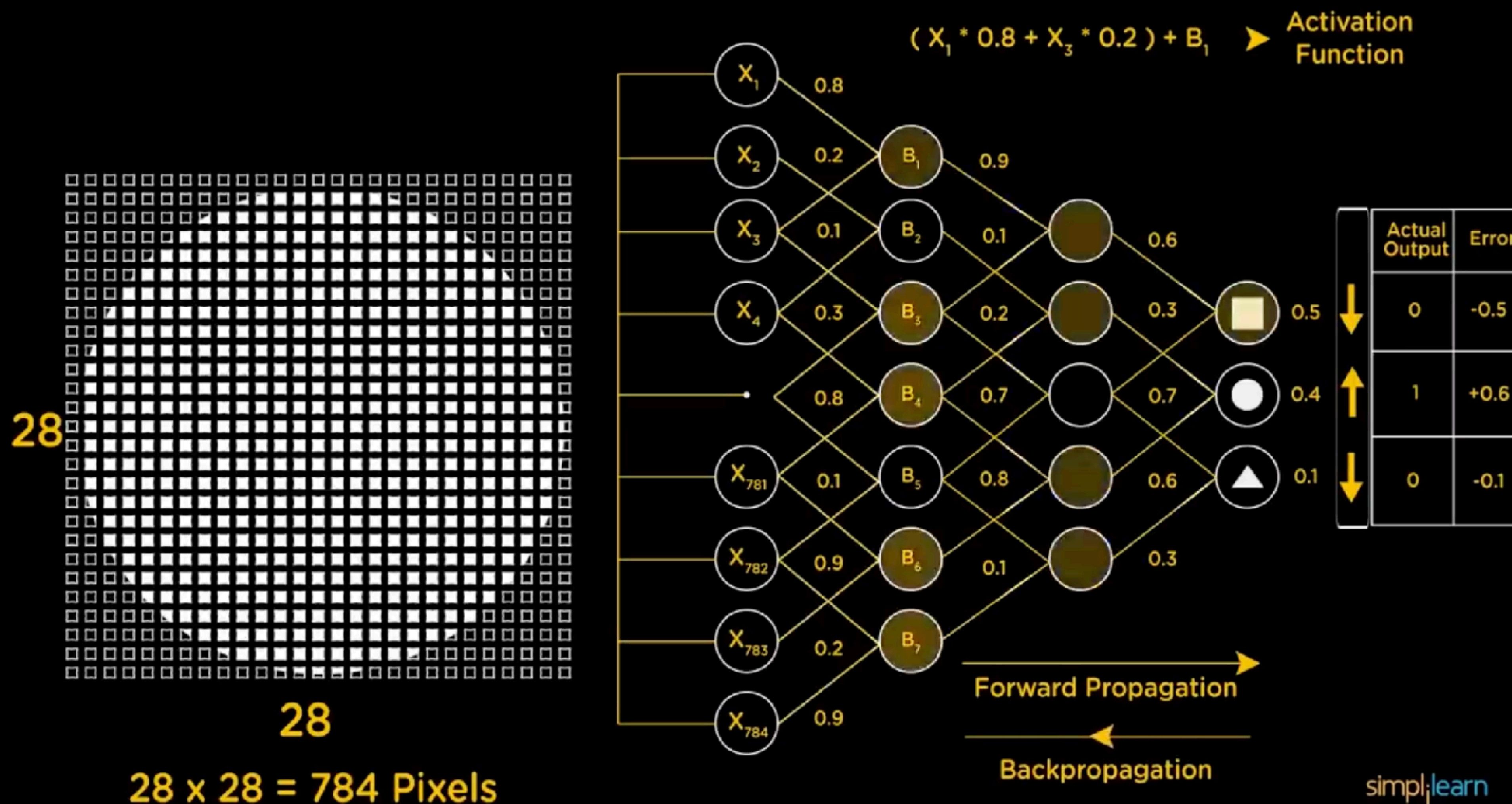
Funcionamiento red “feed forward”



Funcionamiento red “feed forward”



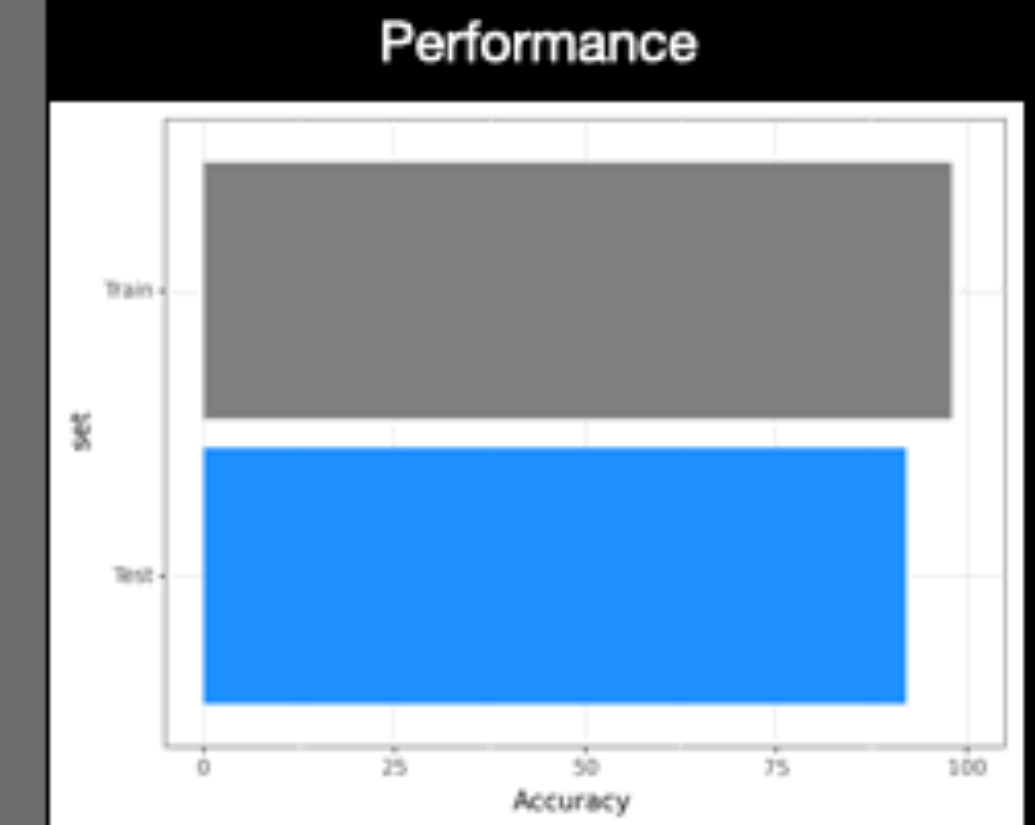
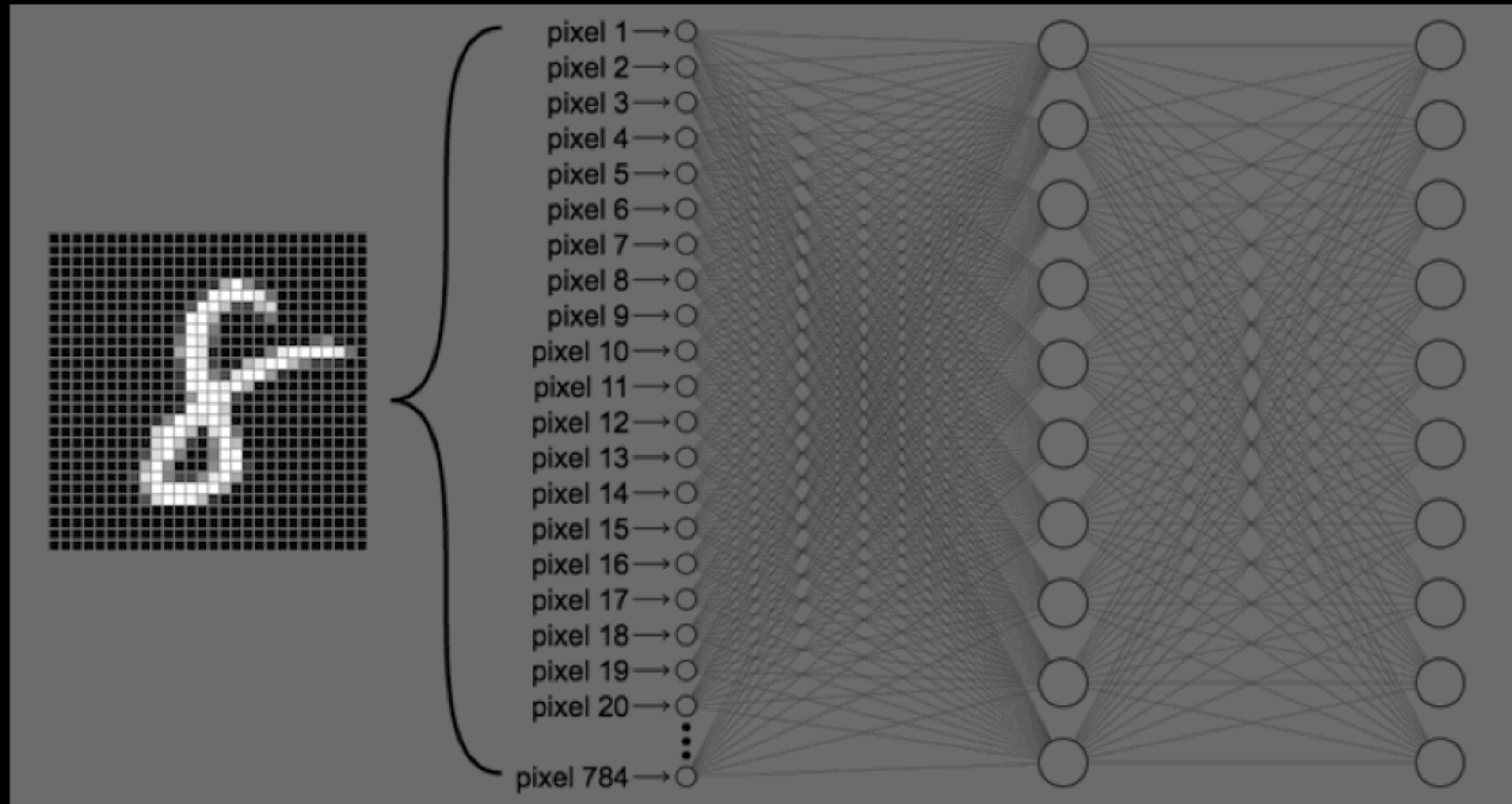
Funcionamiento red “feed forward”



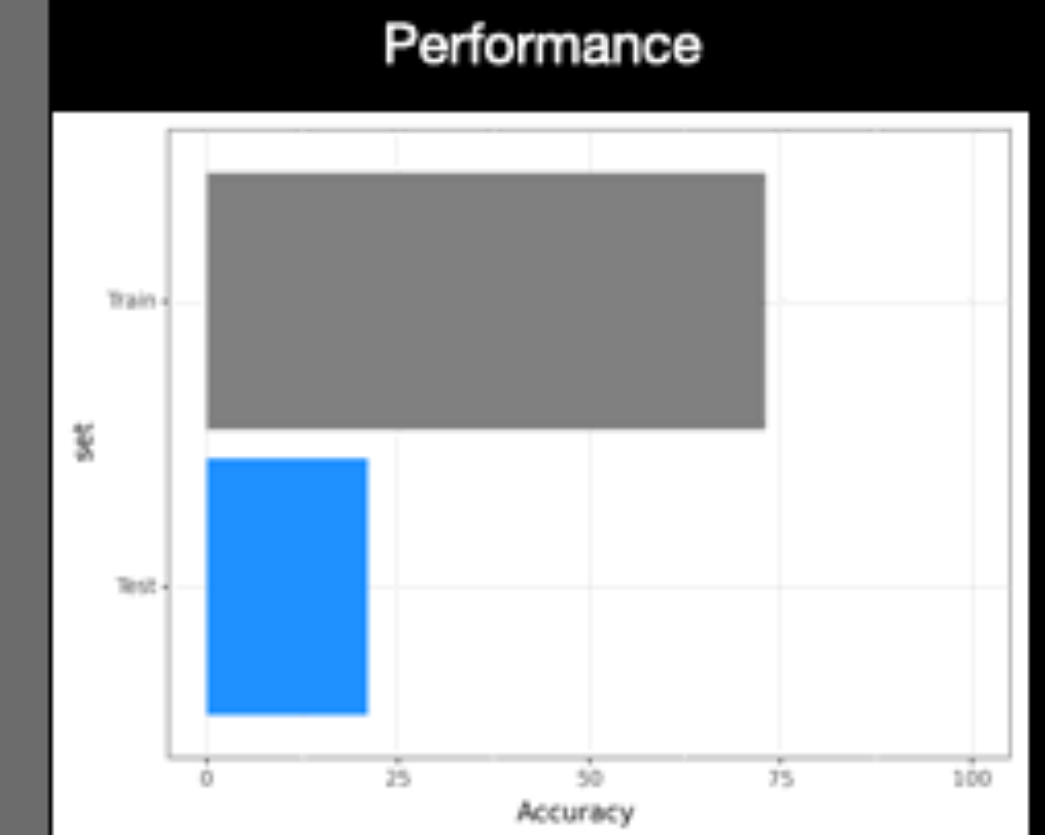
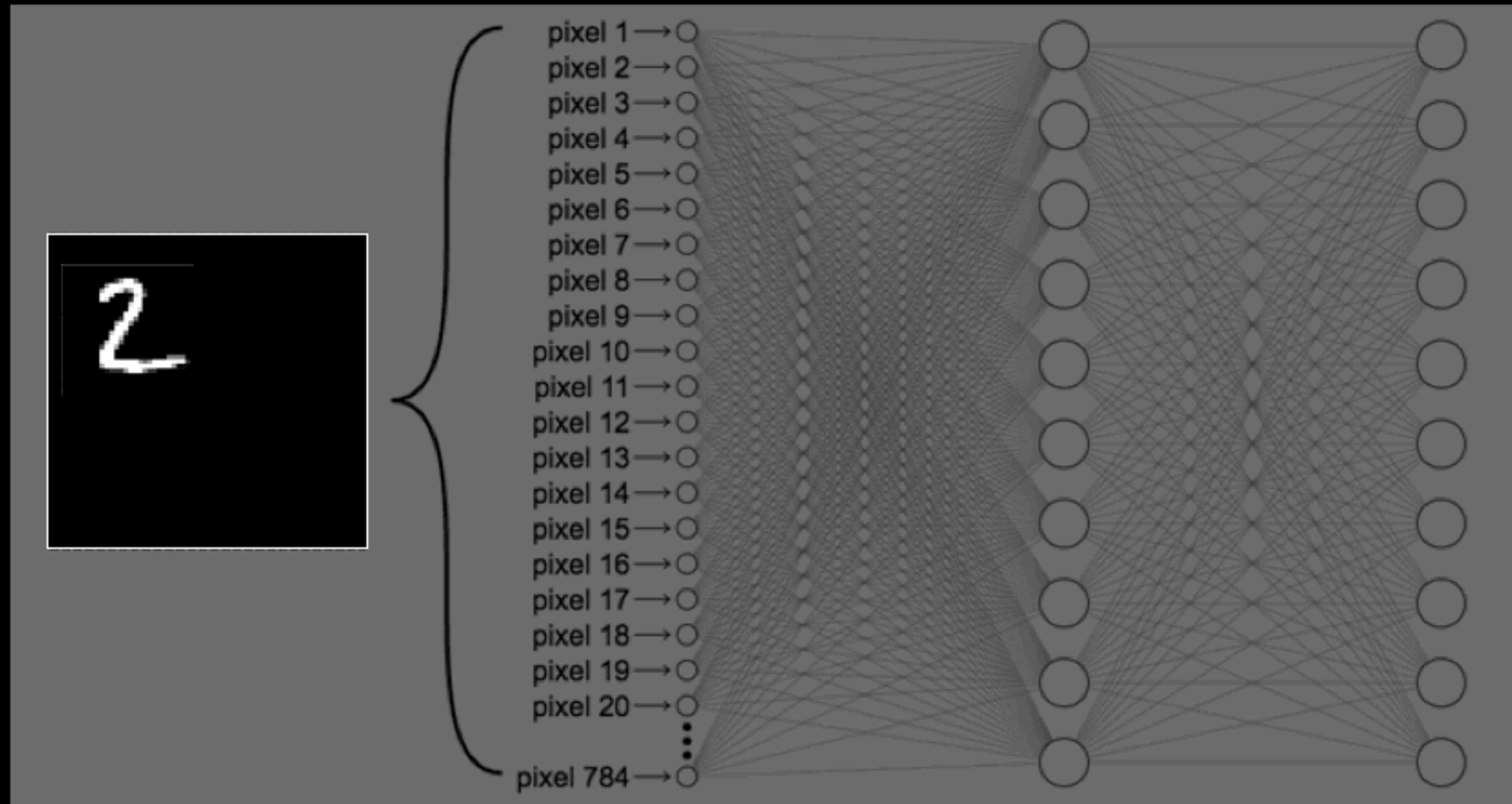


@wmvanvliet

Funcionamiento red “feed forward”



Limitaciones: data del mundo real



Redes neuronales convolucionales (CNN)

- 1980: Neocognitron (Kunihiko Fukushima).
- Inspirada en funcionamiento de sistema visual de vertebrados.

Neocognitron: A Self-organizing Neural Network Model for a Mechanism of Pattern Recognition Unaffected by Shift in Position

Kunihiko Fukushima

NHK Broadcasting Science Research Laboratories, Kinuta, Setagaya, Tokyo, Japan

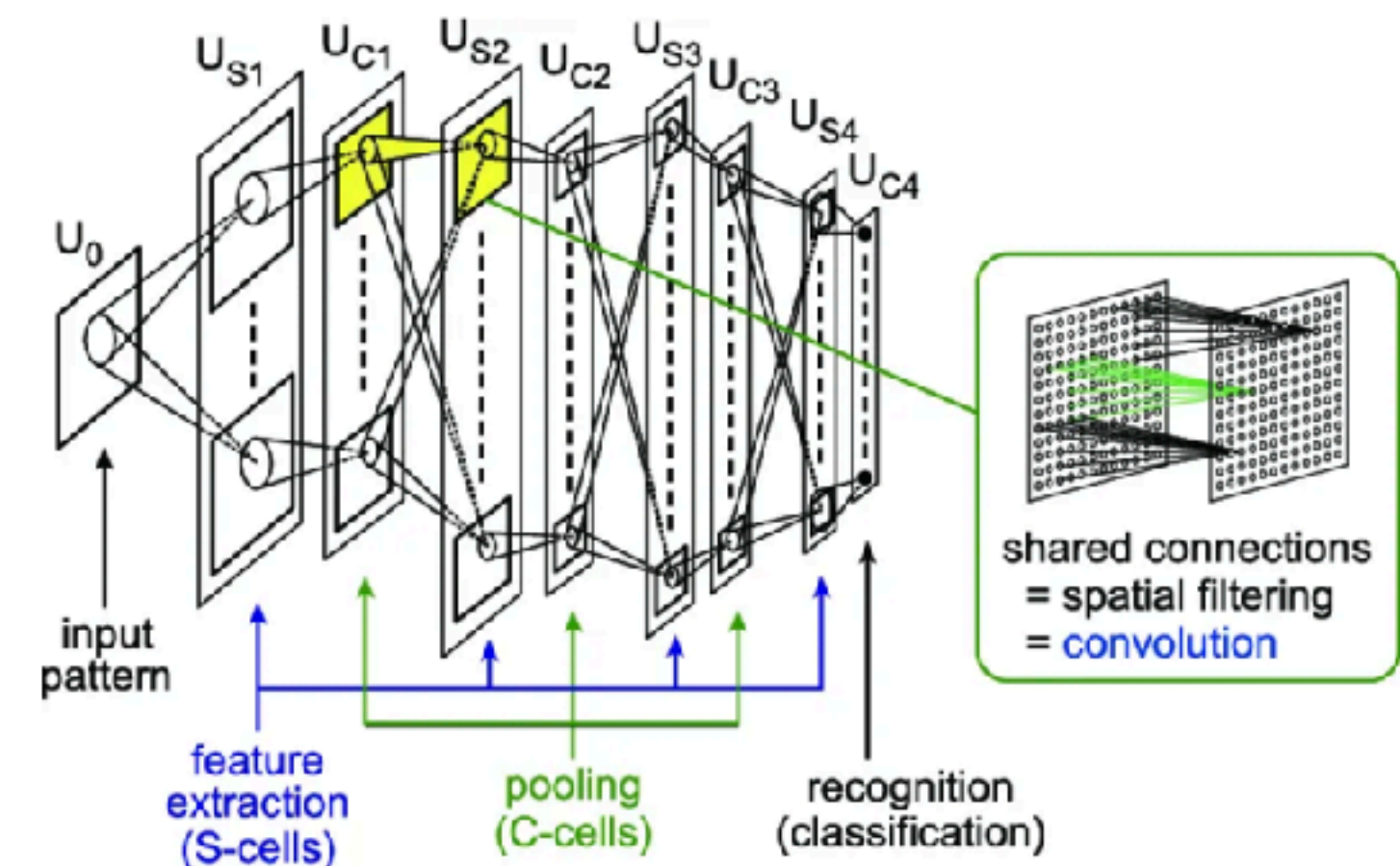


Fig. 1: The architecture of Neocognitron

Redes neuronales convolucionales (CNN)

- 1989 - 1998: LeNet-5 (Yann Lecun).

PROC. OF THE IEEE, NOVEMBER 1998

1

Gradient-Based Learning Applied to Document Recognition

Yann LeCun, Léon Bottou, Yoshua Bengio, and Patrick Haffner

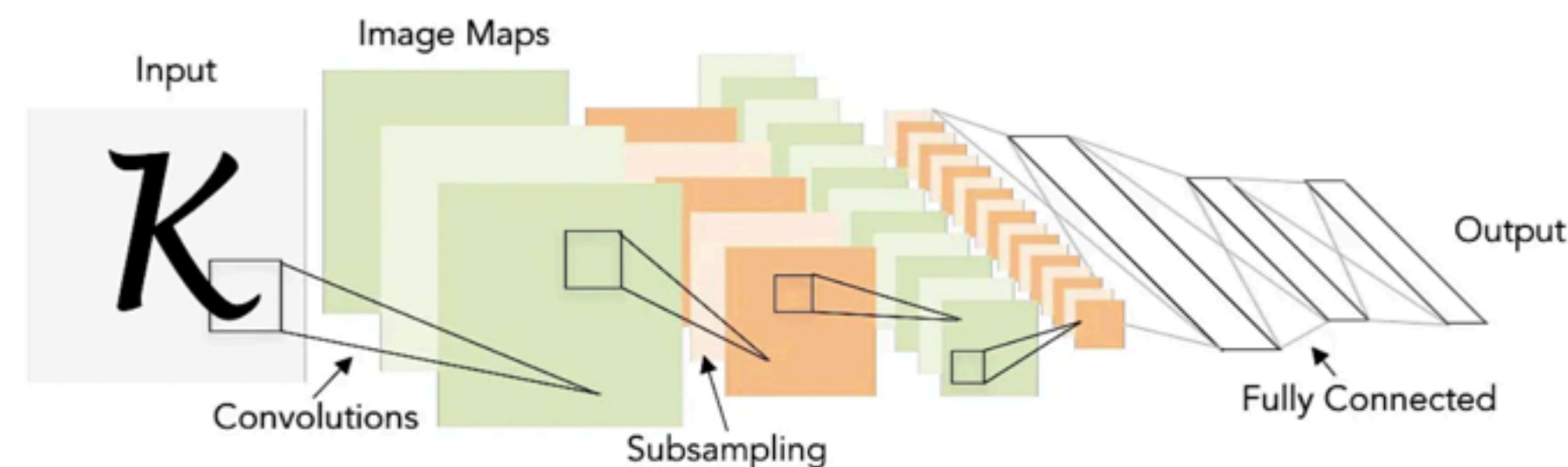
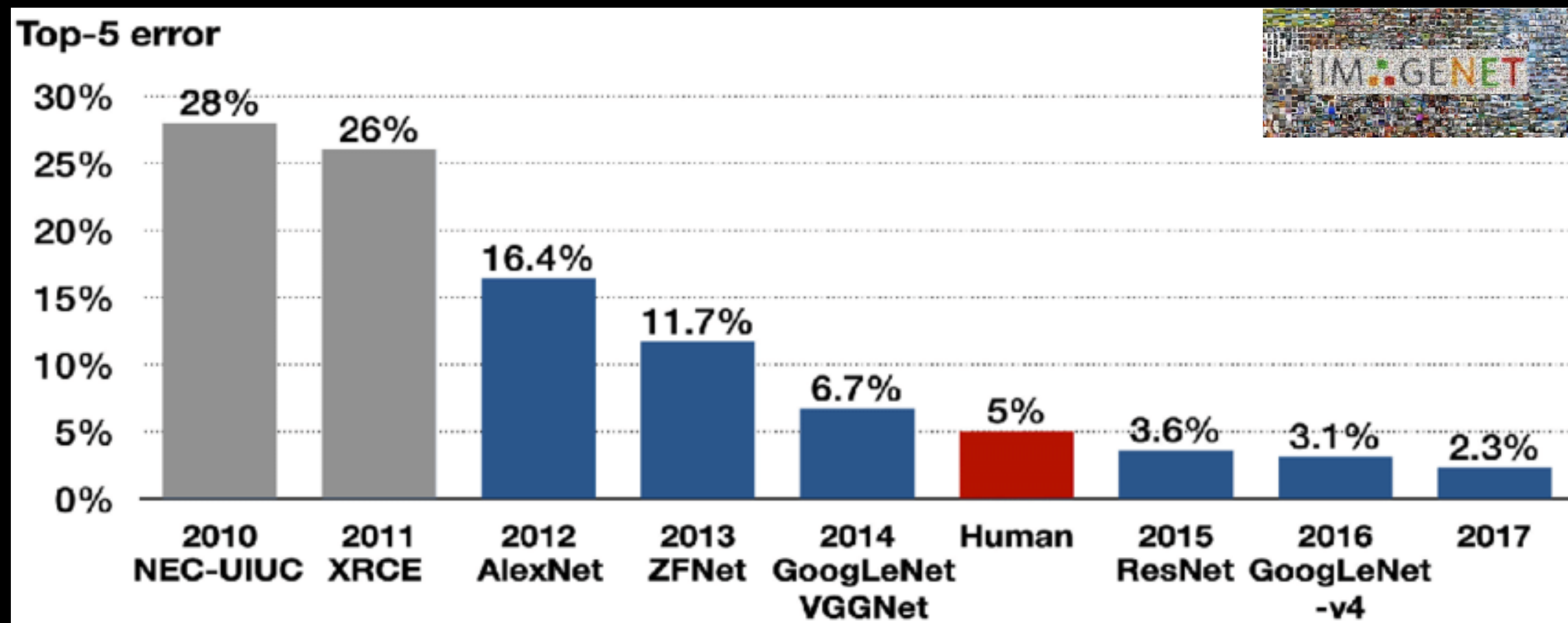


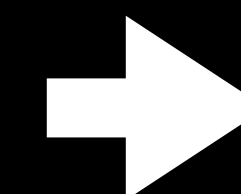
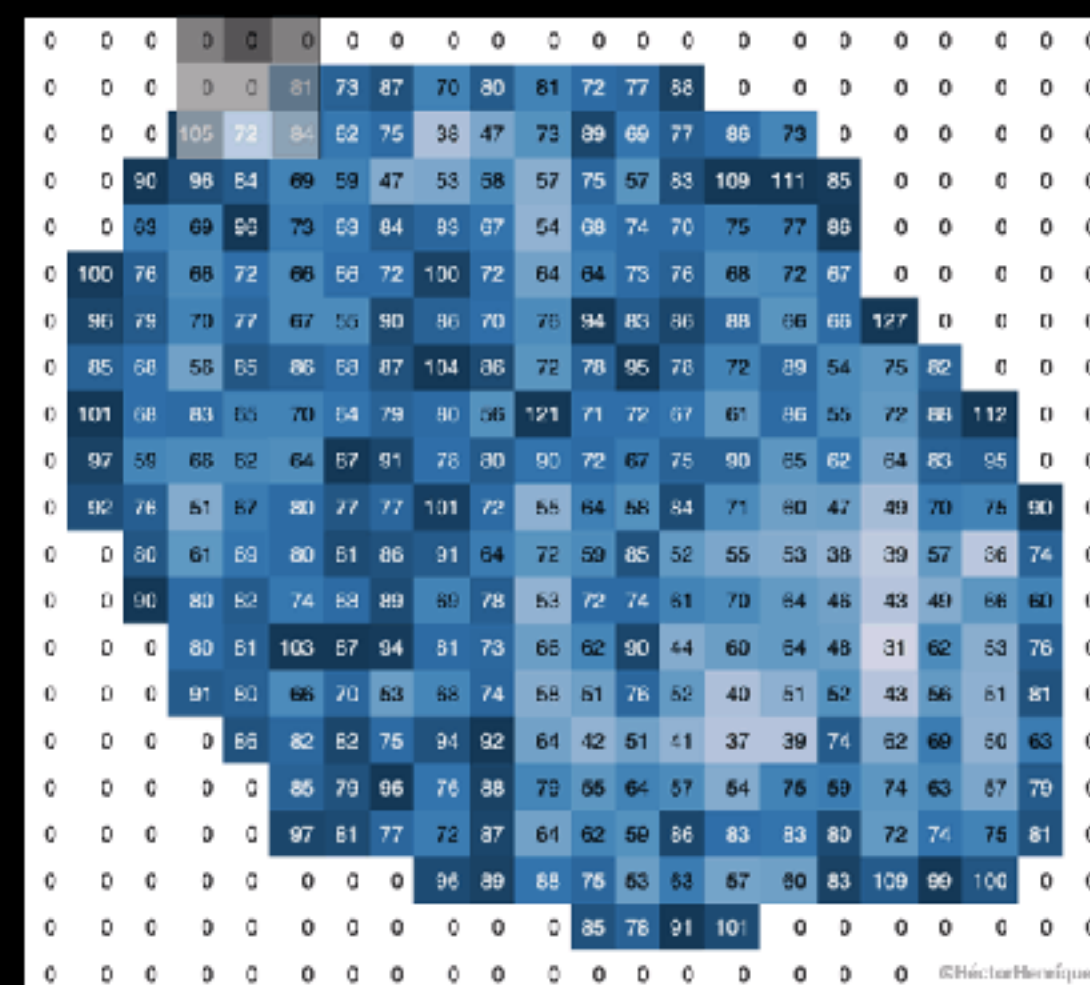
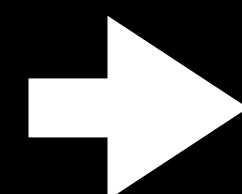
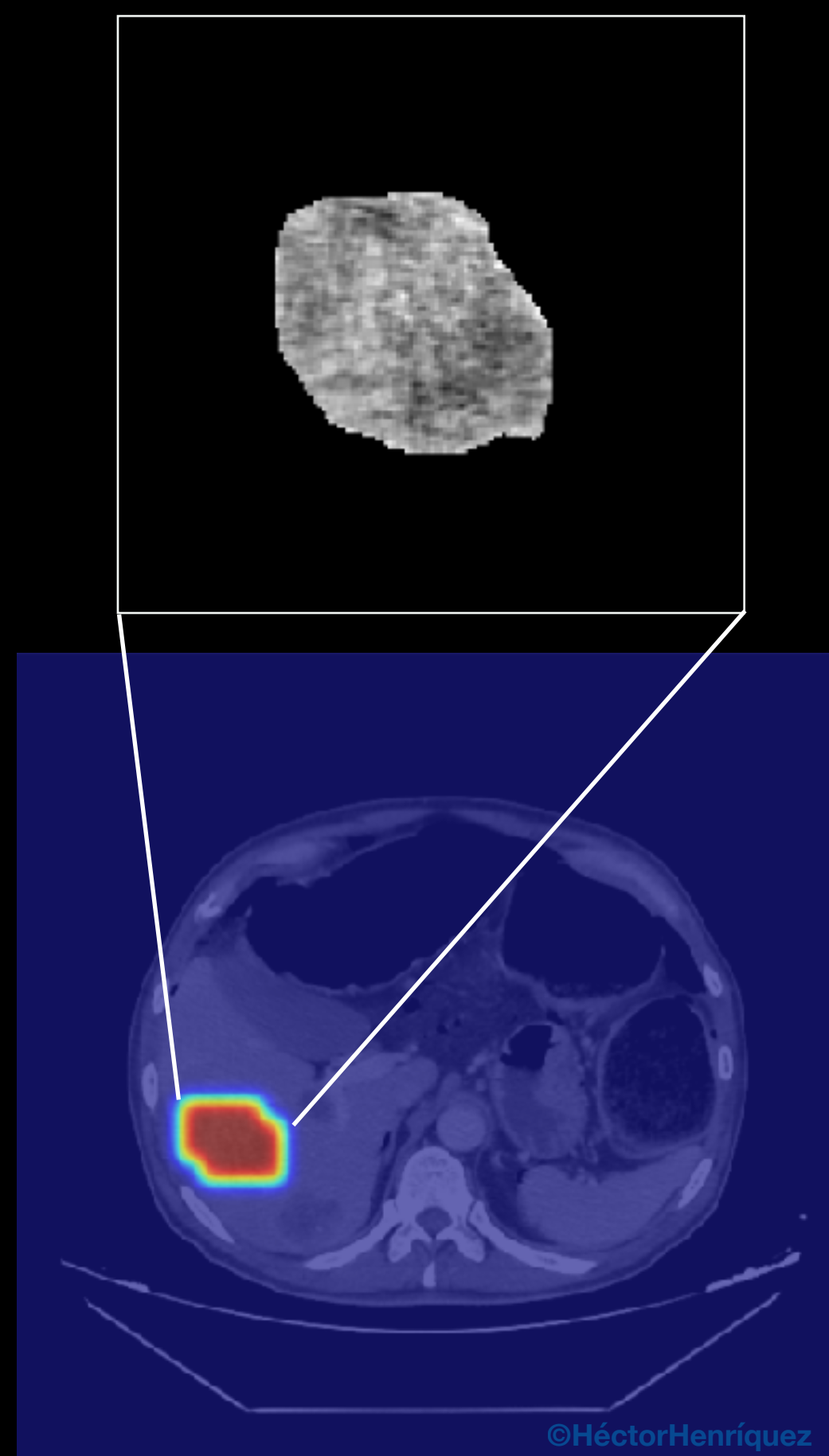
Fig. 2: The architecture of LeNet-5 [7]

Redes neuronales convolucionales (CNN)

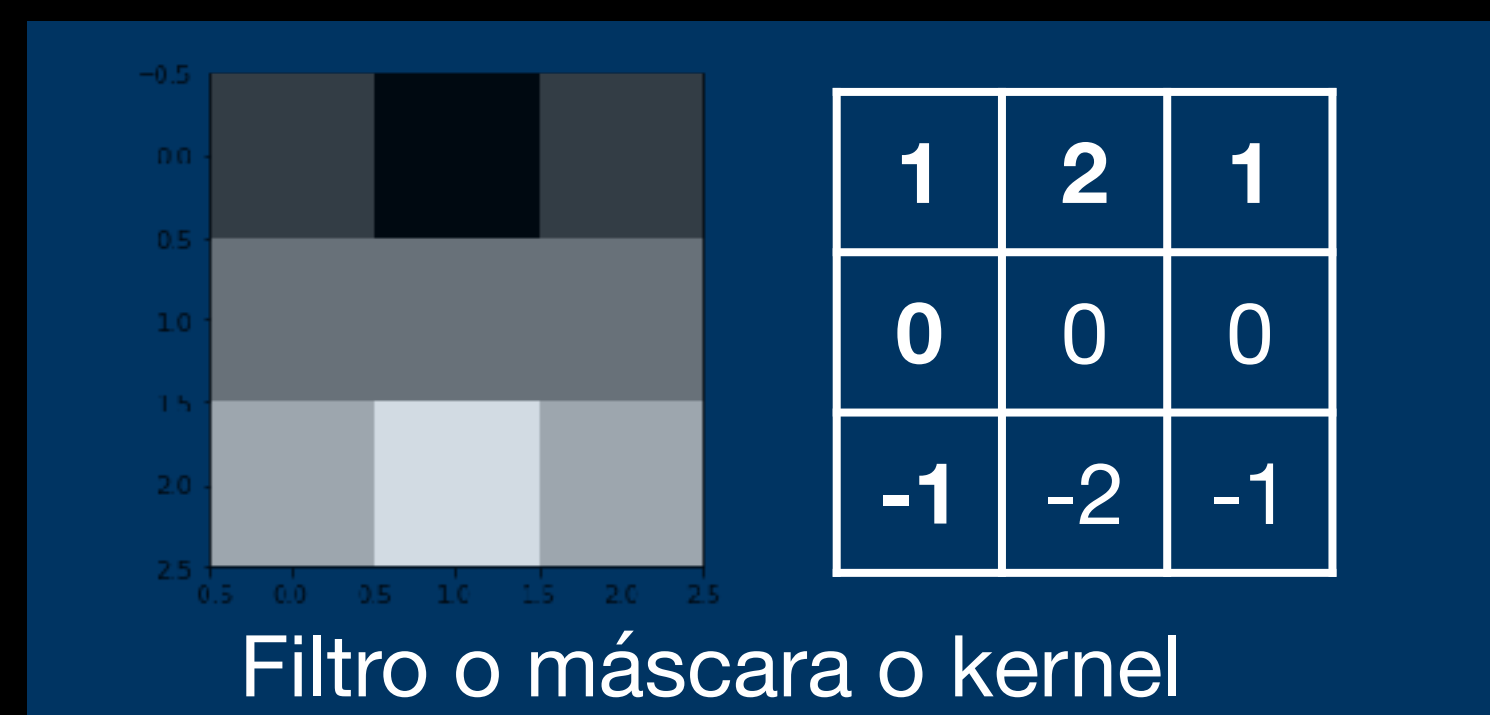
- 2012: AlexNet (Krizhevsky, Sutskever & Hinton).
- Ganadores de ImageNet 2012.



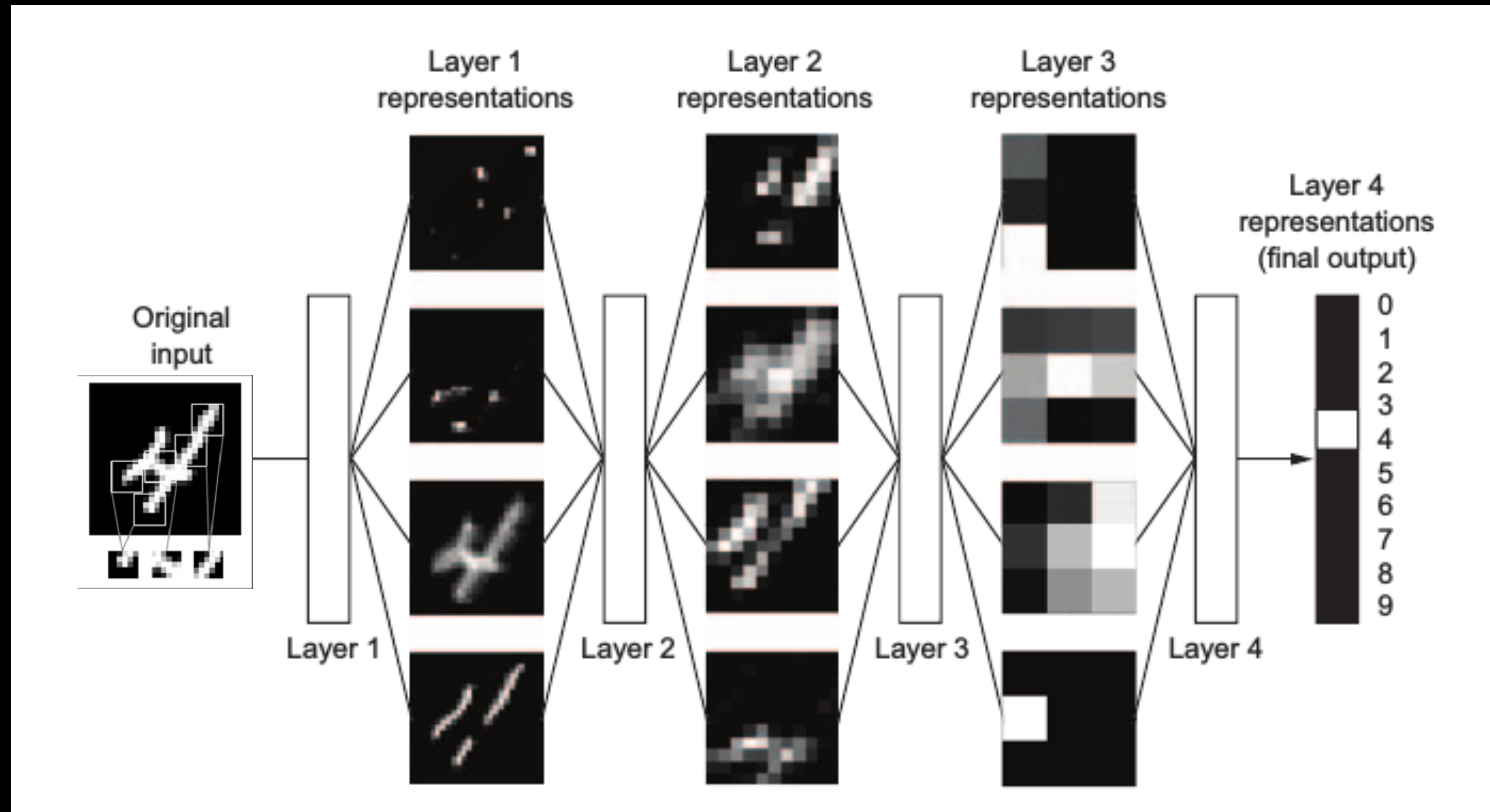
Extracción de características



¿Cuál de estas representaciones es de mayor utilidad?

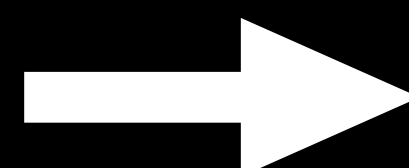


Redes neuronales convolucionales (CNN)

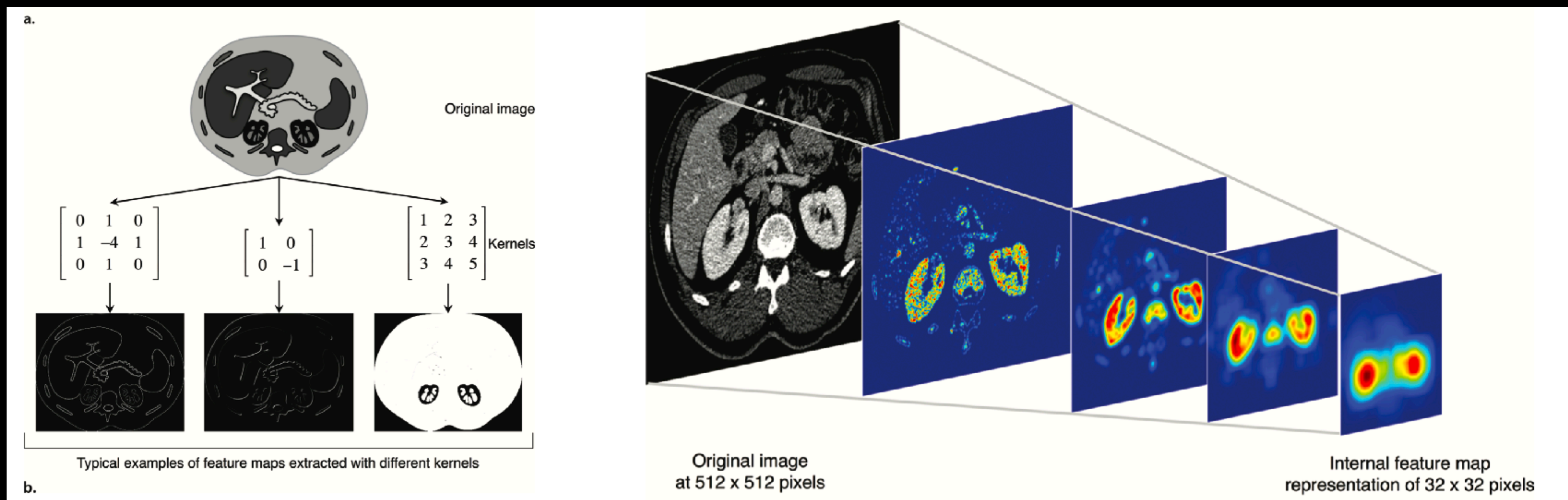


Deep learning e imágenes médicas

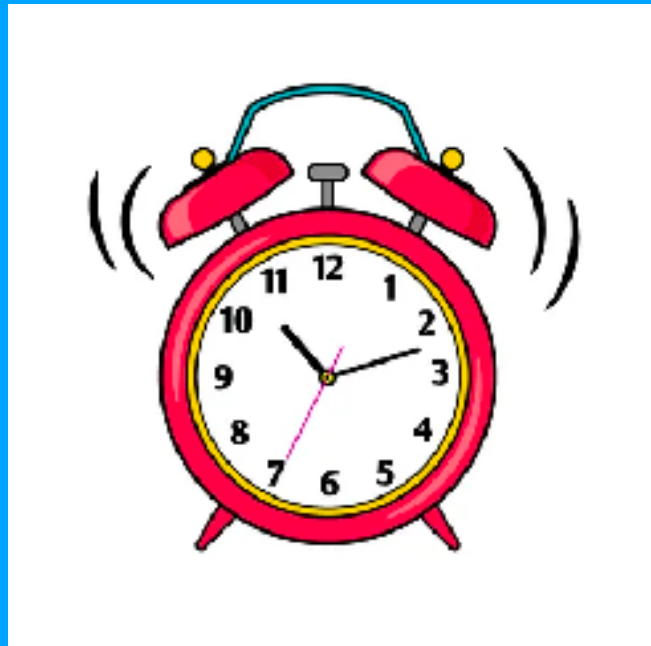
INPUT: Imagen



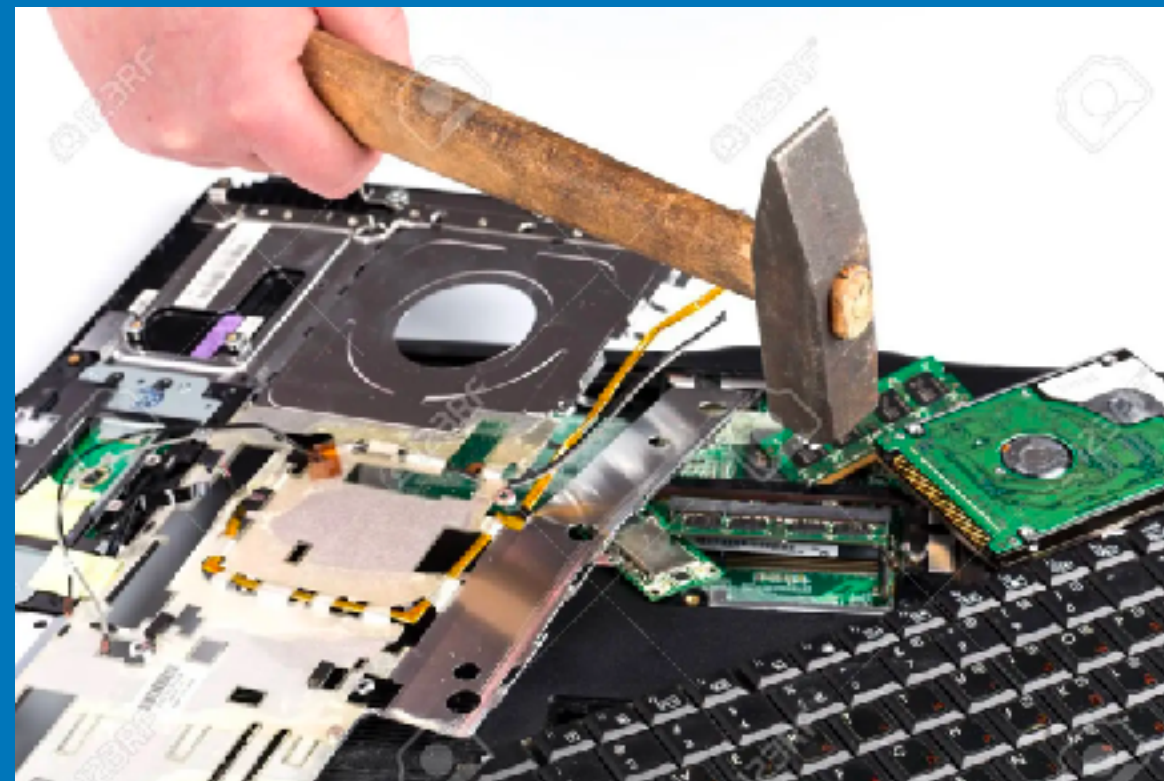
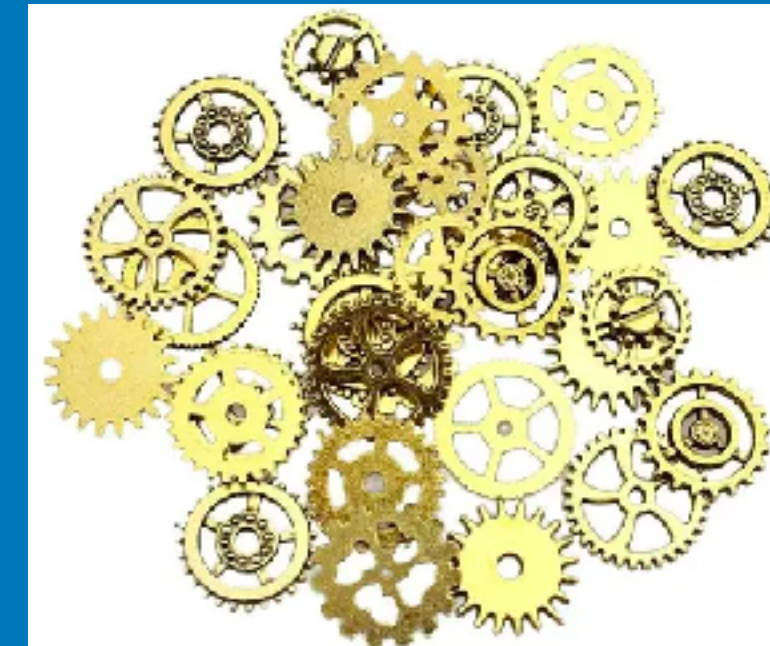
TARGET: "Tumor maligno"



Input



Extracción de características Filtros convolucionales



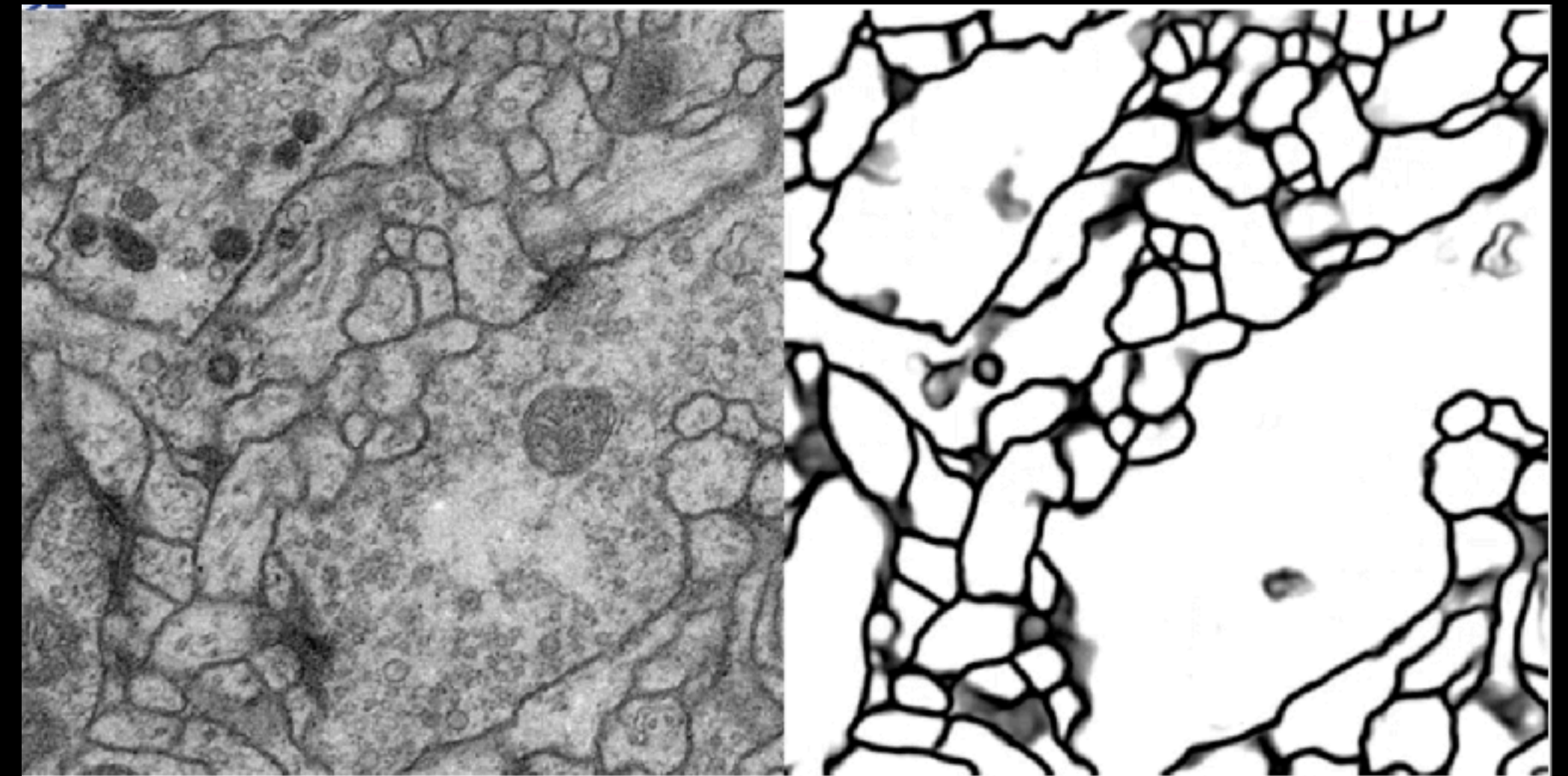
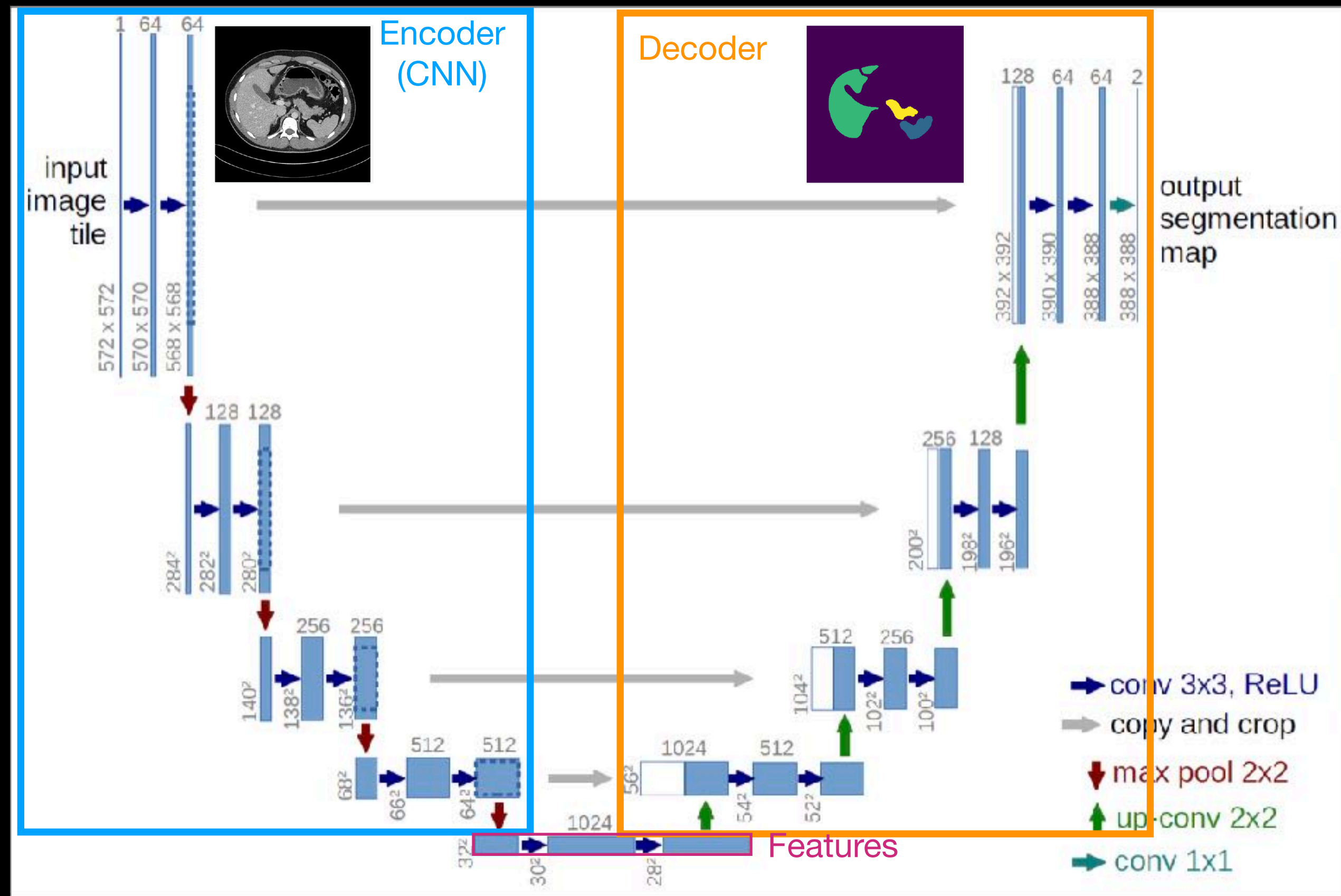
Descomponer, separar y seleccionar

Resumen excesivo

Predicción

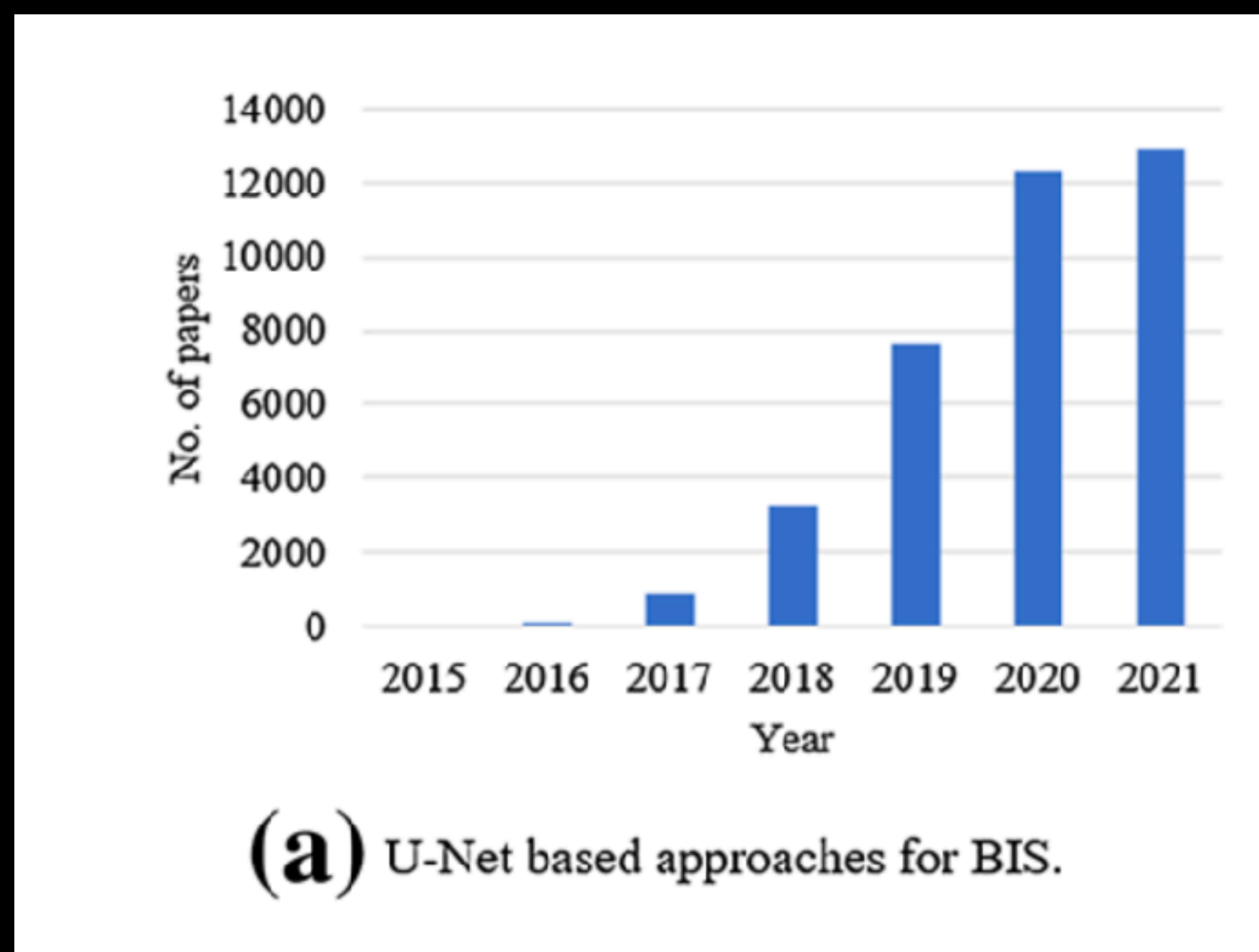


Segmentación de imágenes: Llegada de U-Net



Ronneberger, O., Fischer, P., Brox, T.: U-net: Convolutional networks for biomedical image segmentation. In: Medical Image Computing and Computer-Assisted Intervention–MICCAI 2015: 18th International Conference, Munich, Germany, October 5–9, 2015, Proceedings, Part III 18, pp. 234–241 (2015). Springer

Utilización de U-Net en imágenes médicas

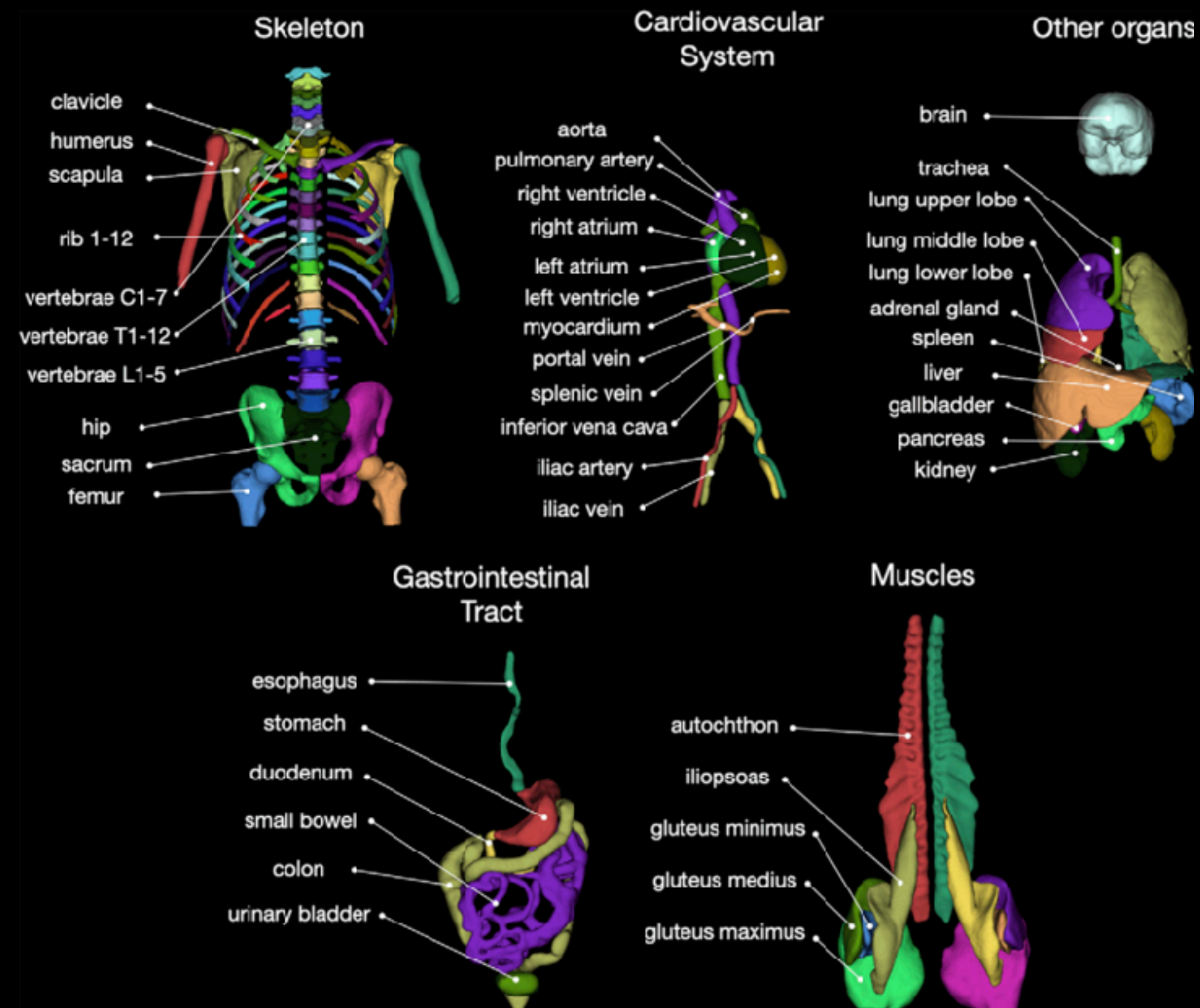


Author	U-Net variant	Modality
Dong et al. (2017)	Modified U-Net	MRI
Rashid et al. (2018)	Modified U-Net	X-ray
Frid-Adar et al. (2018a)	Modified U-Net	X-ray
Que et al. (2018)	CardioXNet framework	X-ray
Oktay et al. (2018)	Attention U-Net	CT
Kohl et al. (2018)	Probabilistic U-Net	CT
Tong et al. (2018)	Improved U-Net	CT
Janssens et al. (2018)	Two stage U-Net model	CT
Kumar et al. (2018)	U-SegNet	MRI
Kermi et al. (2018)	Residual U-Net	MRI
Chen et al. (2018b)	S3DU-Net	MRI
Blanc-Durand et al. (2018)	Vanilla 3D U-Net	PET
Zhao et al. (2018)	3D FCN	PET
Almajalid et al. (2018)	U-Net + SRAD	US
Wang et al. (2018b)	cU-Net	US
Alom et al. (2018)	R2U-Net	Multi-modality
Zhou et al. (2018a)	UNet++	Multi-modality
Subramanian et al. (2019)	CVC framework	X-ray
Li et al. (2019a)	U-Net based framework	X-ray
Dong et al. (2019b)	U-Net-GAN	CT
Liu et al. (2019c)	GIU-Net	CT
Man et al. (2019)	GAU-Net	CT
Seo et al. (2019)	mU-Net	CT
Hiasa et al. (2019)	Bayesian U-Net	CT
Song et al. (2019)	U-NeXt	CT

Siddique, N., Paheding, S., Elkin, C. P., & Devabhaktuni, V. (2021). U-net and its variants for medical image segmentation: A review of theory and applications. *IEEE Access*. <https://doi.org/10.1109/ACCESS.2021.3086020>

Punn, N. S., & Agarwal, S. (2022). Modality specific U-Net variants for biomedical image segmentation: a survey. *Artificial Intelligence Review*, 55(7), 5845–5889. <https://doi.org/10.1007/s10462-022-10152-1>

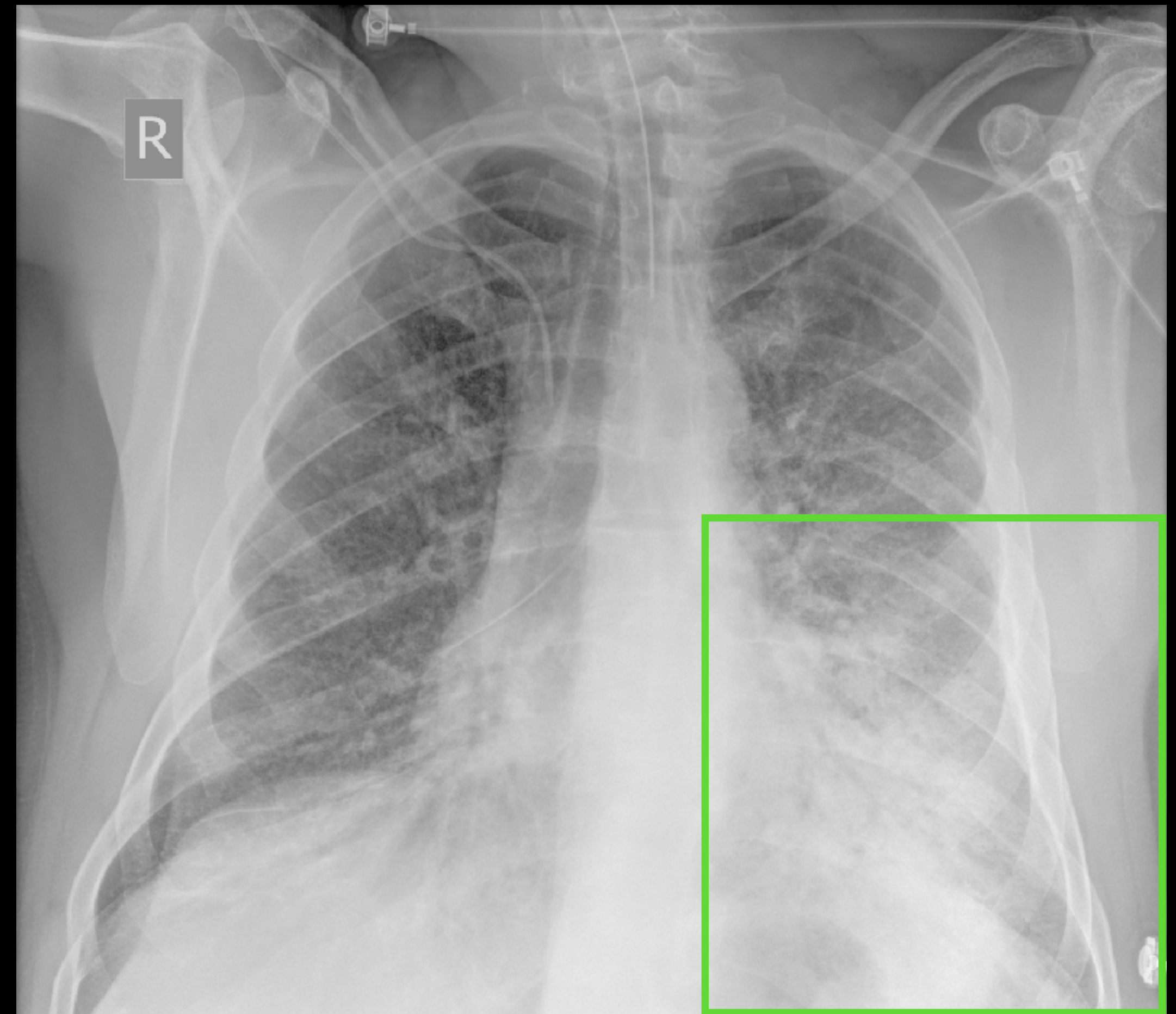
Total Segmentator v2.0



Imágenes Clínica Santa María
Visualización: Héctor Henríquez

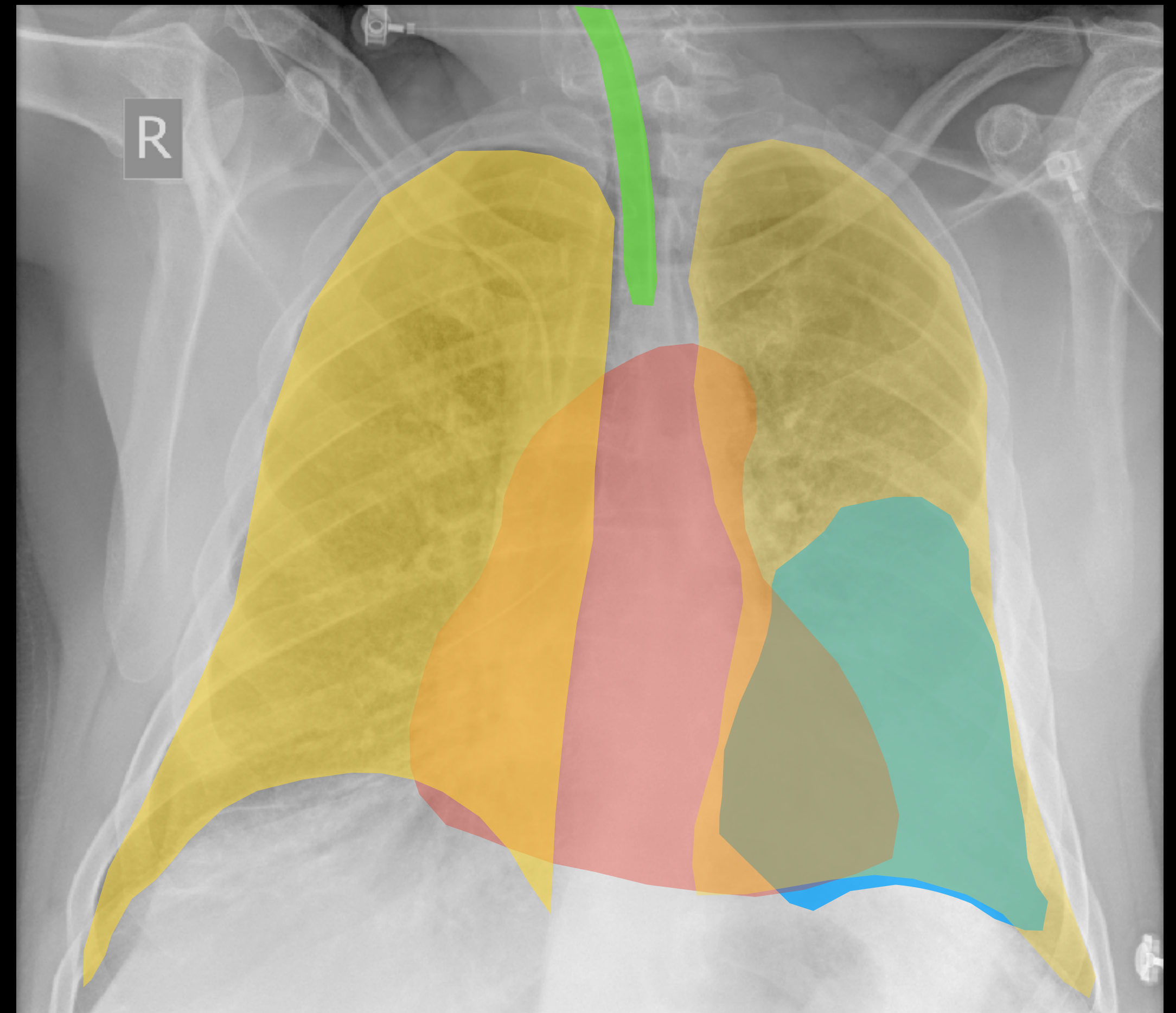
Etiquetado

- Tipos de proyectos:
 - Clasificación de imágenes:
 - Ej. Modelo detección de neumonía.
 - Etiquetas: “neumonía”.
 - Localización de imágenes:
 - Ej. Modelo detección de neumonía.
 - Anotación: Coordenadas del hallazgo etiquetado como neumonía.



Etiquetado

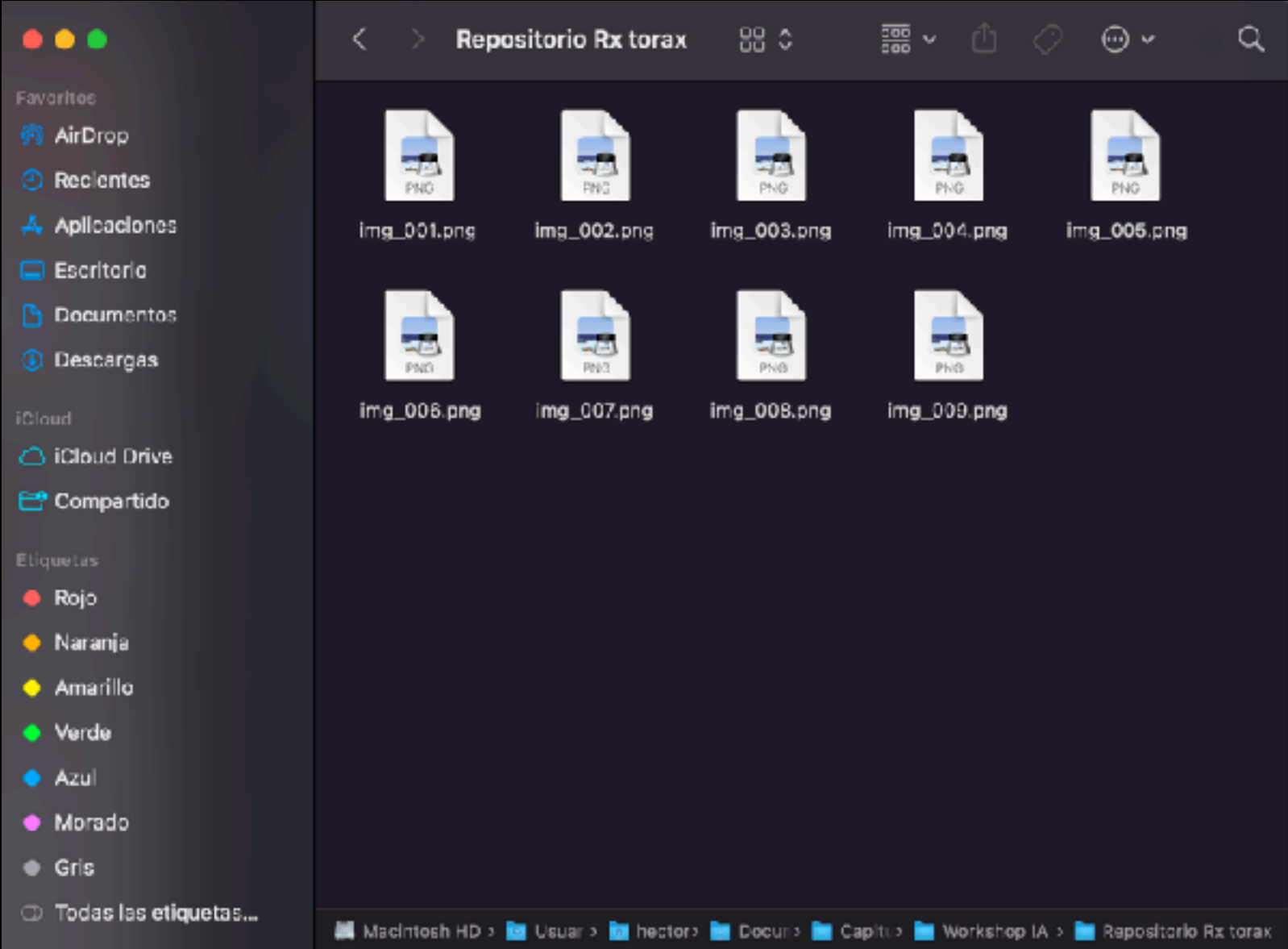
- Tipos de proyectos:
 - Segmentación de imágenes
 - Ej. Modelo detección de neumonía.
 - Anotación:
 - Imagen: Máscara de segmentación.
 - Segmentación pulmonar (0)
 - Segmentación neumonía (1)
 - Segmentación corazón (2)
 - Segmentación tubo endotraqueal (3).



Flujo de trabajo

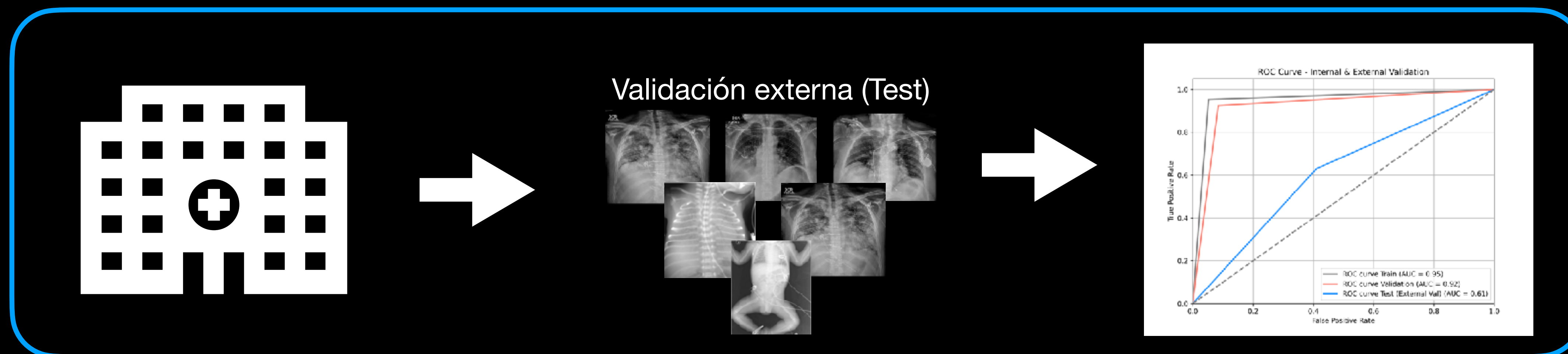
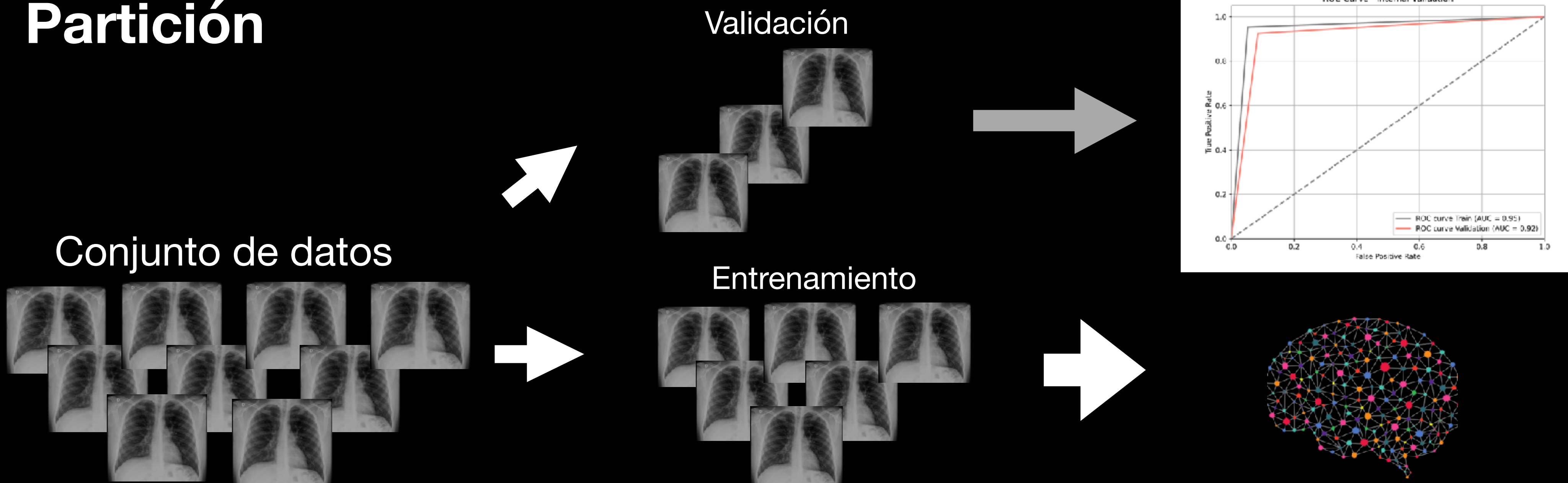
Set de datos

ID paciente	Género	Edad	id_imagen	Target
id_001	M	34	img_001	Neumonía
id_002	F	25	img_002	Normal
id_003	F	68	img_003	Normal
id_004	M	58	img_004	Neumonía
id_005	M	32	Img_005	Normal
id_006	F	18	Img_006	Normal
id_007	F	67	Img_007	Neumonía
id_008	F	83	img_008	Normal
id_009	M	47	img_009	Neumonía



Flujo de trabajo

Partición



Flujo de trabajo

Partición

ID paciente	Género	Edad	id_imagen	Target	Set
id_001	M	34	img_001	Neumonía	Train
id_002	F	25	img_002	Normal	Test
id_003	F	68	img_003	Normal	Train
id_004	M	58	img_004	Neumonía	Test
id_005	M	32	img_005	Normal	Train
id_006	F	18	img_006	Normal	Test
id_007	F	67	img_007	Neumonía	Train
id_008	F	83	img_008	Normal	Train
id_009	M	47	img_009	Neumonía	Train

Flujo de trabajo

Partición

Train
Test
Train
Train
Test
Train
Test
Train
Train

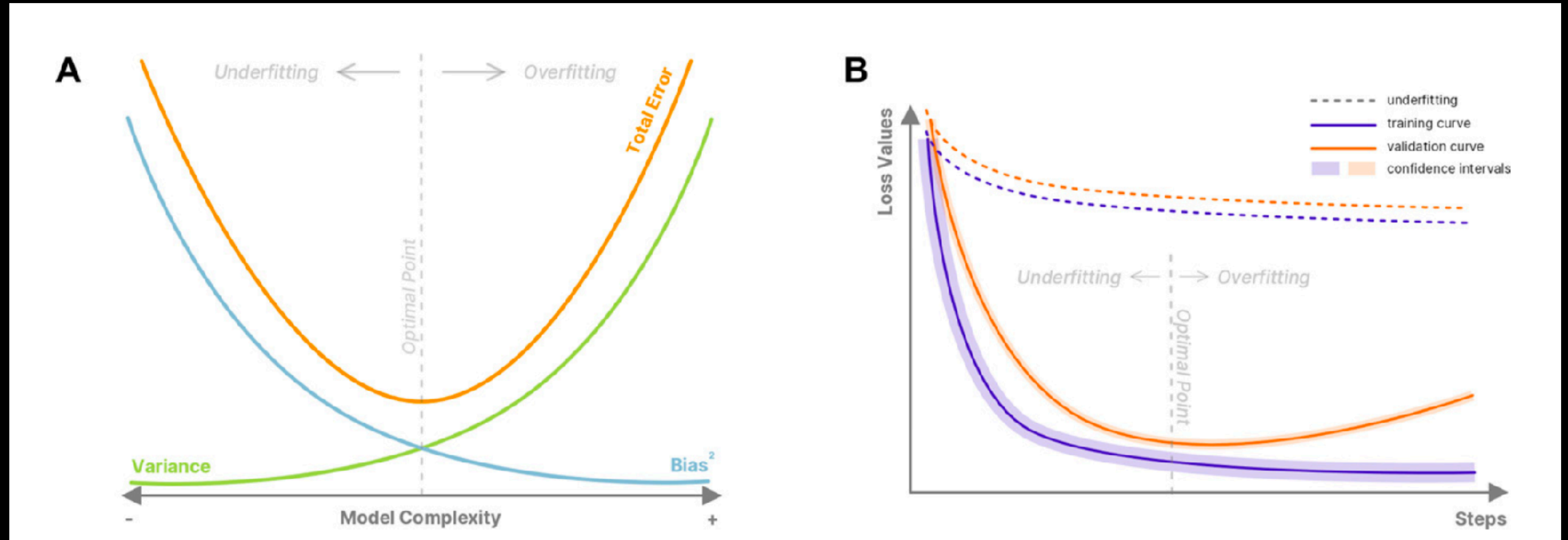
- “Hold out”:
 - 10/90
 - 20/80
 - 30/70

Test	Train	Train	Train	Train
Train	Test	Train	Train	Train
Train	Train	Test	Train	Train
Train	Train	Train	Test	Train
Train	Train	Train	Train	Test

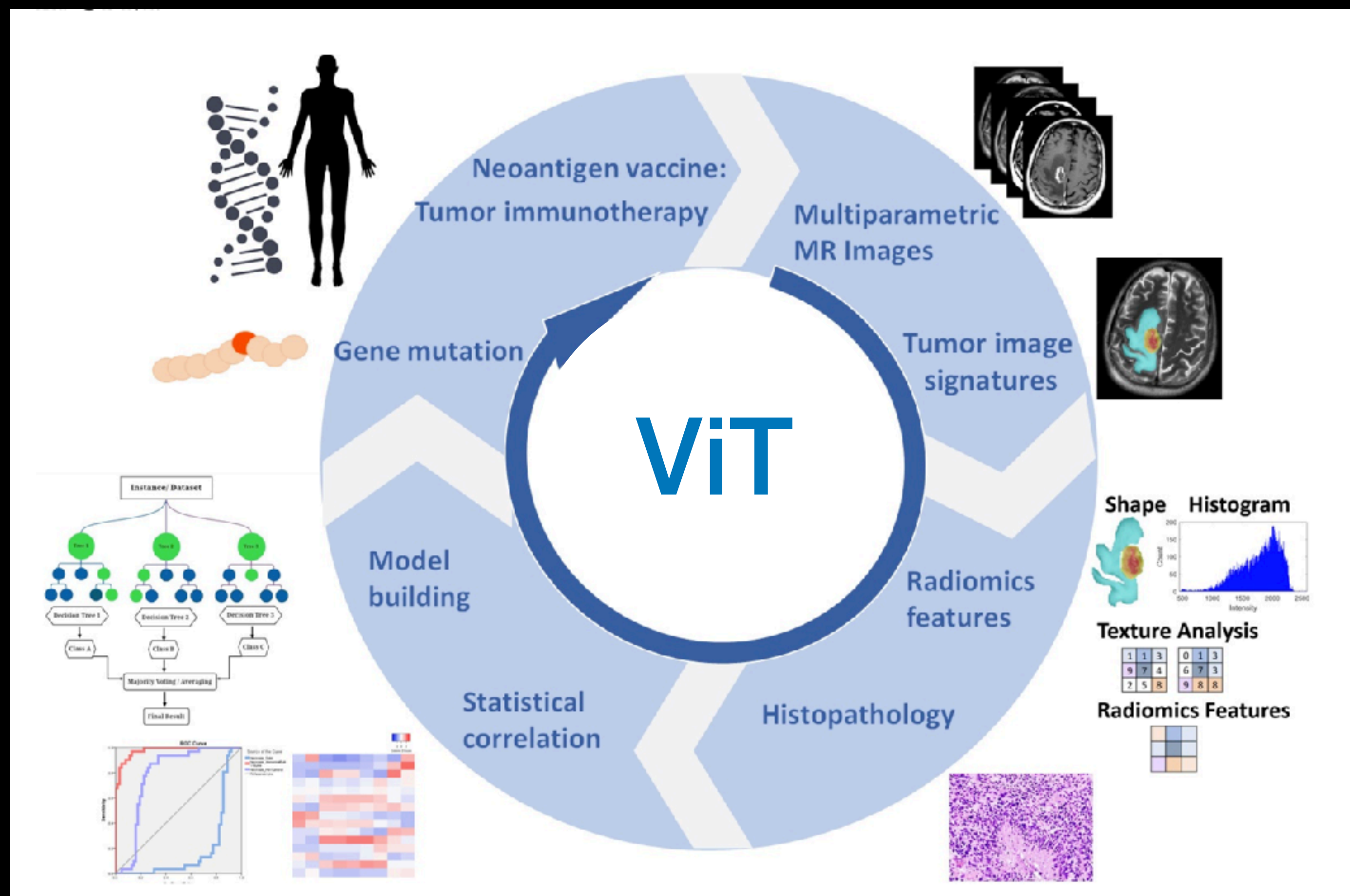
- K-fold Cross Validation:
 - $K = 5 - 10$

Flujo de trabajo

Partición & Evaluación



Mensaje final



Chiu, F. Y., & Yen, Y. (2022). Efficient Radiomics-Based Classification of Multi-Parametric MR Images to Identify Volumetric Habitats and Signatures in Glioblastoma: A Machine Learning Approach. *Cancers*, 14(6). <https://doi.org/10.3390/cancers14061475>